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Insights into ontology-based model-based systems engineering: state of the art and enabling framework

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A B S T R A C T

Ontology offers a formalized framework for knowledge representation and reasoning in complex systems, enabling the development of model-based systems engineering (MBSE). This study conducts a systematic bibliometric analysis of 566 papers on ontology in MBSE from 2008 to 2024, complemented by a qualitative literature review. The findings reveal a marked upward trajectory in research activities, characterized by small-scale collaboration networks predominantly led by core scholars. The research topics converge on several key topics related to the subject of ontology in MBSE, including modeling language, decision-making and knowledge management, life cycle integration, architecture modeling, and interoperability. From the perspective of these key topics, in this study, a systematic literature review was conducted to propose an ontology-based MBSE enabling framework. This framework consists of: (1) an ontology-based MBSE paradigm that reconfigures MBSE processes through enhanced semantic interoperability, optimized life cycle management, and dynamic decision support mechanisms; (2) a reference architecture that integrates MBSE models with ontology; and (3) key technologies that address semantic consistency verification and multidomain integration. Finally, the industrial applications are discussed as illustrative, preliminary, and qualitative evidence, suggesting the potential applicability of the proposed ontology-based MBSE framework in real-world engineering scenarios, particularly in the aviation industry. This work offers a structured overview of the current state, research hotspots, future trajectories, and enabling framework of ontology in MBSE. Future work can further investigate standardized evaluation metrics and benchmarking methods to quantitatively assess the effectiveness and performance of the OBMBSE framework in industrial applications.

1. Introduction

With the ongoing technological evolution of Information and Communications Technology (ICT), industrial systems have become increasingly complex due to intensified interactions among heterogeneous mechatronic components and embedded software [1]. The interactive coupling and interdependence of system components and processes enable interconnected and integrated functionalities in industrial systems, while simultaneously posing significant challenges for effective industrial information integration and management [2]. Traditional system design paradigms, which rely heavily on static electronic documentation, often lead to fragmented, inconsistent, and siloed information, hindering effective communication and collaboration among stakeholders and across different life cycle stages [3]. Model-based systems engineering (MBSE) introduces a transformative shift from static documents to dynamic and interconnected models [4], which has been increasingly adopted in industry to support industrial information integration engineering [5] and is rapidly becoming a sine qua non in industry [6].

By leveraging digital models that formalize system requirements, design, analysis, verification, and validation activities [7], MBSE enhances information consistency and traceability throughout the

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engineering system life cycle. This type of consistency and traceability is fundamentally dependent on preserving semantic consistency across all life cycle models. These models are constructed using formal modeling languages [8], whose semantics provide the essential foundation for the precise and unambiguous representation of different system viewpoints, including structure, behavior, constraints, and interactions [9]. To achieve a model-based representation of complex engineering systems from different system engineering viewpoints, MBSE relies on a variety of modeling languages [10], including the Systems Modeling Language (SysML) [11], Object-Process Language (OPL) [12], Business Process Model and Notation (BPMN) [13], and Architecture Analysis and Design Language (AADL) [10]. However, each of these modeling languages has its own semantics and syntax [14], that are tailored to specific engineering domains and life cycle stages [15]. Existing model transformation techniques can address syntactic inconsistencies, but they do not resolve semantic misalignment across modeling languages. As a result, semantic heterogeneity arises in system models, which substantially increases cognitive load and interpretation costs [16].

With the introduction of the semantic web in computer science in the early 2000s, ontologies, as a “semantic cornerstone” of the semantic web, have garnered significant attention for addressing semantic heterogeneity in industrial domains[17]. Ontologies ensure consistent interpretation of terms and concepts across engineering domains by structuring knowledge into well-defined hierarchies, relationships, and formalized semantic constraints [18,19]. This provides a standardized conceptual framework that enables the alignment of multiple modeling languages, mitigates semantic inconsistencies, and facilitates seamless model exchange and integration across tools and domains [20]. For instance, Lu et al. [21] demonstrated that the ontology developed using the Graph, Object, Property, Point, Relationship, Role, Extension (GOPPRR-E) method could support the formalism and semantic interoperability of at least five MBSE modeling languages, including SysML, BPMN, OPL, Unified Profile for DoDAF/MODAF (UPDM), and Electronics Architecture and Software Technology-Architecture Description Language (EAST-ADL). Furthermore, grounded in formal logic (e.g., first-order logic and description logic), ontologies enable automated reasoning mechanisms to verify the logical consistency and semantic completeness of models, mitigating design conflicts in the early design phase and reducing design rework. Implicit knowledge can be automatically inferred, and in turn, decision-making is facilitated in engineering system design by leveraging axioms and rules within an ontology.

Recently, several reviews related to ontologies and MBSE have been published, as summarized in Table 1. Most of these reviews characterize ontologies as a supporting technology for addressing specific MBSE scenarios, such as change analysis, modeling language integration, or formal modeling. The authors have highlighted the fact that ontologies could improve MBSE model standardization, reusability, consistency, and tool interoperability. However, these reviews lacked a systematic analysis of the general MBSE scenarios that ontologies can support, as well as a detailed discussion of the enabling technologies that underpin the effective application of ontologies in MBSE. Moreover, some reviews have noted that applying ontology in MBSE is a future trend, but they do not explain what this trend entails in concrete terms. Thus, current research faces two main research gaps:

(1) Existing literature does not provide a systematic and concrete analysis that addresses the state of the art and emerging trends of applying ontologies in MBSE. This hinders researchers from both obtaining a systematic understanding of this field and clearly identifying promising directions for future research;

(2) Existing work neither systematically discusses related conceptual definitions to clarify the scope of applying ontologies in MBSE nor provides the reference architectures for applying ontologies across MBSE scenarios and enabling technologies required for their effective application. This severely limits the understanding of the theoretical foundations and practical applications of ontologies in MBSE, further hindering the advancement of research in this field.

To address these gaps, this study conducts a systematic review of ontology in MBSE using a bibliometric analysis [26] to analyze the state of the art and emerging trends of ontology in MBSE. Based on these insights, alongside a systematic literature review [27], this study proposes an enabling framework to systematically address the related conceptual definition, reference architecture, and enabling technologies. The main contributions of this work can be summarized as follows:

Table 1
Previous reviews related to ontology in MBSE.

Authors	Scope	Method	Main findings	Limitations
Akundi et al. [7]	2,380 MBSE-related publications (1995–2020)	Text mining	Ontology in MBSE for change analysis is a representative focus within the topic “MBSE for requirements specification, representation, and analysis.”	Focuses on the applications of ontology in MBSE for the specific research topic “change analysis.”
Li et al. [22]	4675 MBSE-related publications (1993–2022)	Bibliometric analysis	Unifying multidomain modeling languages through ontologies is necessary in current MBSE practice; research on ontology in MBSE represents a future direction.	Focuses on the applications of ontologies in a specific MBSE scenario, “modeling language integration”, but it remains unclear which specific research avenues are encompassed by “ontology in MBSE” as a future research direction.
Dong et al. [23]	143 MBSE-related publications (up to 2022)	Word cloud analysis	The application of ontology in MBSE is emerging; building ontology to formalize relevant conceptual models and improve the standardization and reusability of MBSE models is expected to be an important research direction for MBSE.	Focuses on the specific research topic and direction, “formal modeling” for ontology in MBSE.
Yang et al. [19]	116 OBSE-related publications (up to 2019)	SLR	Ontology in MBSE plays a key role in consistency management and interoperability, facilitates communication among stakeholders by integrating standard UML/SysML *meta*-models with domain ontologies, and supports the development of modeling languages through formal semantics and logical mechanisms.	Lacks the enabling technologies that underpin the effective application of ontologies in MBSE, and focuses on the application of ontologies in a specific MBSE scenario, “modeling language integration.”
Bolshakov et al. [24]	108 publications on digital transformation and MBSE (2018–2022)	Inductive approach	Ontology in MBSE for formal modeling is recognized as a recommended principle.	Focuses on the application of ontologies in a specific MBSE scenario, “formal modeling.”
Velthuizen et al. [25]	1563 MBSE-related publications (2007–2025)	Bibliometric analysis	Building standardized ontologies for MBSE to support a shared understanding, knowledge transfer, and interoperability across tools, disciplines, and organizations is a future research direction.	Remains unclear which specific research avenues are encompassed by “building standardized ontologies for MBSE” as a future research direction.

(1) A structured analysis of the research landscape of ontology in MBSE is conducted through bibliometric analysis, including research teams, publication sources, current research topics, and future trends. Insights derived from the analysis of current research topics are further leveraged to guide the systematic literature review (SLR), thereby enabling a focused, coherent, and in-depth examination of the field.

(2) Based on the SLR, an ontology-based MBSE (OBMBSE) enabling framework is proposed, which comprises a conceptual definition of OBMBSE, a reference architecture for integrating ontology with MBSE, and key enabling technologies. The proposed framework establishes a solid theoretical foundation and provides a systematic pathway for the practical application of OBMBSE in engineering projects.

Furthermore, this study discusses selected industrial applications of the OBMBSE framework in real-world engineering scenarios to contextualize its potential relevance to industrial practice.

The rest of this article is organized as follows. Section 2 presents the research methodology. The results of the bibliometric analysis are presented in Section 3. Section 4 introduces the enabling framework for OBMBSE and several industrial applications. Finally, conclusions are discussed in Section 5.

2. Research methodology

The research methodology of this study involves four steps as shown in Fig. 1: (1) data collection, (2) data processing, (3) data analysis, and (4) systematic analysis. The details are as follows:

2.1. Data collection

For this step, we collected publications related to ontology and MBSE up to 2024, with the search conducted on October 21, 2024. Web of Science (WoS), Scopus, and Google Scholar are widely recognized as major sources of publication metadata and impact metrics [28]. Among these sources, Google Scholar provides the broadest coverage. However, it was excluded from the research due to the uncertain quality of some indexed resources and the lack of transparent inclusion criteria [28]. Moreover, to ensure consistency and reproducibility in the bibliometric analysis while minimizing the risk of duplicate records, a single, well-established database was selected. Compared to WoS, Scopus provides broader and more comprehensive coverage of relevant literature. Moreover, the availability of individual profiles for all authors, institutions, and serial sources, as well as the interconnected database interface, makes Scopus more user-friendly for practical use. Therefore, Scopus was selected as the sole data source for this study. In addition, to ensure conceptual consistency and alignment with the core research focus on "ontology in MBSE," the search strategy was deliberately restricted to the keywords "MBSE" and "ontology."

The search strategy was as follows:

LANGUAGE (English) AND ALL (MBSE) AND ALL (ontology)

The LANGUAGE (English) parameter restricts the search results to English-language publications only. This is because English is the dominant language for disseminating high-impact and peer-reviewed research in the field of MBSE and ontology. The search parameter ALL indicates a search across all fields, including the title, abstract, keywords, and other metadata. ALL (MBSE) and ALL (ontology) refer to retrieving the literature containing the terms "MBSE" and "ontology" across all fields. AND is a logical operator, meaning both conditions must be met. Finally, 704 publications were compiled and recorded in both .csv and .bib file formats in accordance with the Scopus information standard.

2.2. Data processing

The collected data were standardized to ensure the dataset's quality, consistency, and relevance, thereby enhancing the accuracy and reliability of the insights. The final bibliometric data were obtained through the following filtering steps:

(1) Removed publications: Scopus includes various types of publications. First, the conference reviews, errata, and repeated publications were removed from the dataset. Second, it was found that the abbreviation MBSE did not always refer to *model-based systems engineering*, so the publications that included other uses of the term, such as *missing blank-space errors* or *model-based software engineering* were deleted. Third, publications that mentioned *MBSE* or *ontology* in the references but were not relevant to the topic were excluded.

(2) Cleaning the dataset for consistency in keywords: To ensure consistency in handling large datasets, terminology was standardized by unifying variations in acronyms, singular or plural forms, hyphenation, and redundant annotations. Examples include consolidating "model-based systems engineering" into "MBSE," harmonizing variants of "cyber physical system" and "cyber-physical systems" into "cyber-physical system", and eliminating "(MBSE)" from "MBSE (MBSE)." This

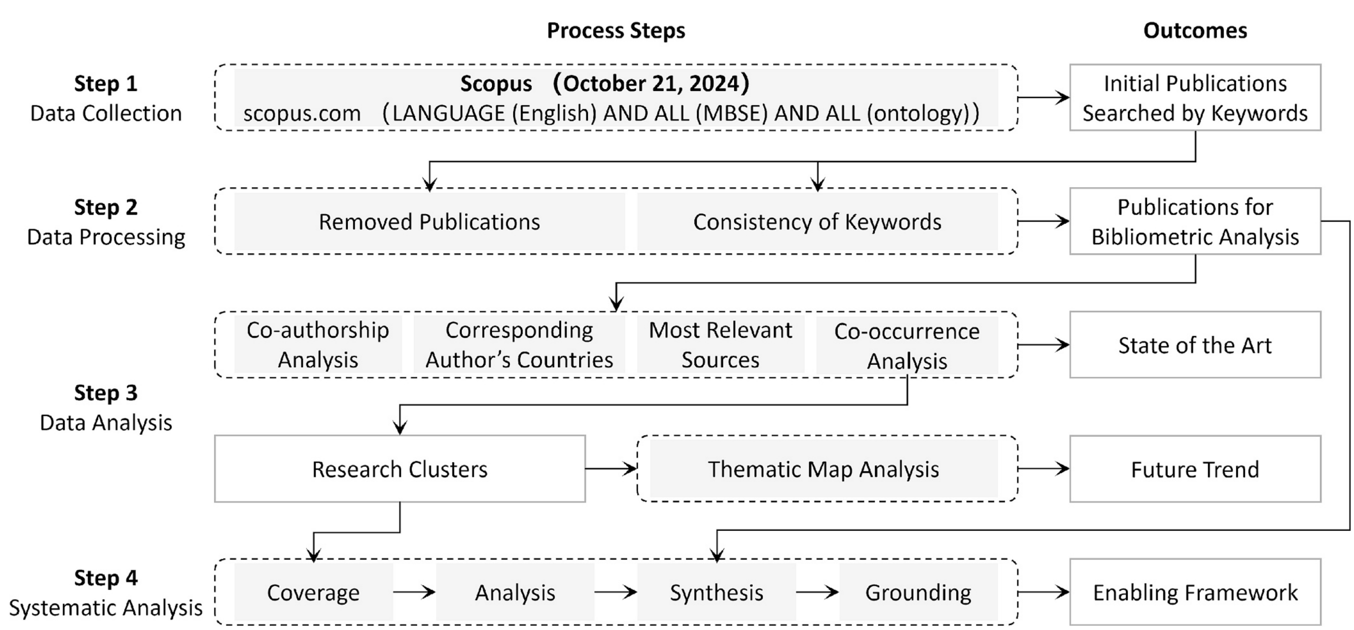

Fig. 1. Research methodology.

process eliminated ambiguity and ensured uniform keywords for visualization. Finally, 566 publications were retained for data analysis.

2.3. Data analysis

To analyze the collected dataset, this study employed two widely recognized bibliometric analysis tools, VOSViewer and R (BibliometriX package) [29], owing to their complementary capabilities in visualizing and statistically analyzing bibliometric data. The following analyses were conducted:

(1) Co-authorship analysis: Co-authorship is one of the most verifiable methods used to quantify or analyze scientific collaboration [30]. The minimum number of publications per author was set to three to ensure a clear and interpretable visualization. This analysis revealed the structure of research teams and their collaboration patterns, which helped identify potential research topics and trends in ontology and MBSE.

(2) Corresponding author's countries: The countries of corresponding authors provided insights into the global distribution of research leadership and collaboration patterns. Focusing on the top 20 countries, this analysis not only reflected the intensity and structure of international scientific collaboration but also highlighted broader policy directions and strategic priorities in the research on ontology in MBSE across both academia and industry.

(3) Most relevant sources: This analysis highlighted key publications central to research on ontology in MBSE, reflecting the main research domains in the field. It also provided insights into potential research topics and emerging trends. By examining the top 10 sources, we identified the most influential publications and their contributions to advancing the research on ontology in MBSE.

(4) Co-occurrence analysis: Frequent keywords and their relationships were visualized using VOSViewer, revealing research clusters in the field of ontology in MBSE and highlighting current research topics. To ensure clear visualization and meaningful results, the minimum threshold for keyword occurrence was set to 10. Insights from this analysis helped identify critical topics for ontology in MBSE, which directly informed the SLR and guided the development of the enabling framework proposed based on the SLR.

(5) Thematic map: Based on the co-occurrence analysis, the centrality and density were used to classify the research clusters of ontology in MBSE, providing insights into current themes and future research trends. High-centrality and high-density clusters indicated well-developed, mainstream topics, whereas low-centrality or low-density clusters denoted emerging or peripheral research areas. This analysis provided an overview of the development of ontology in MBSE and helped distinguish between core and emerging technologies in the field.

2.4. Systematic analysis

Building on the broad landscape identified by the preceding bibliometric analysis, a systematic literature review (SLR) was conducted to provide a focused synthesis of the research on ontology in MBSE. This provided a foundation for developing an enabling framework in this field. The SLR process is illustrated in Fig. 1. This process adhered to the procedure described in Reference [31].

(1) Coverage: Coverage delimited the precise scope of literature pertinent to research on ontology in MBSE. This boundary was established through key search terms and defined inclusion criteria. Selecting appropriate search terms denoted a critical step in the SLR process, ensuring the retrieval of academic papers most relevant to the field. Derived from the thematic clusters identified through co-occurrence analysis of the existing literature on ontology in MBSE, these specific cluster topics served as the foundational search terms for this review. Based on these key search terms, selection criteria were formulated to filter existing studies, as shown in Table 2. Specifically, the acceptance criteria were applied conjunctively, meaning a study was retained only if it met all required conditions simultaneously. Conversely, the exclusion criteria were applied disjunctively, such that a study was excluded if it met any one of the exclusion criteria. Finally, the full texts were meticulously reviewed to ensure their relevance. Following this screening procedure, 52 primary studies were finally retained and analyzed in Section 4.1.

Table 2
Selection criteria for systematic literature review on ontology in MBSE.

Inclusion Criteria	Exclusion Criteria
The title, abstract, keywords of a paper include at least one of the search terms.	No search terms are included in the title, abstract, and keywords of a paper.
The title, abstract, keywords of the paper must include "ontology/ontologies" and "MBSE/SysML (which is the most common modeling language)."	The title, abstract, and keywords do not explicitly contain the terms "ontology" or "ontologies."
	The title, abstract, and keywords do not explicitly contain "MBSE" or "SysML."

(2) Analysis: The analysis phase involved a systematic and rigorous examination of the selected publications to extract, categorize, and document their individual contributions to the research domain of ontology in MBSE. The selected 52 primary studies were analyzed in Table 4 (Section 4.1), which serves as the foundation for the development of an enabling framework in the synthesis process.

(3) Synthesis: The synthesis stage of the literature represented the author's original contribution to the field. Up until this stage, 52 publications on ontology in MBSE were collected, and it was stated what each individual publication says (analysis), without explaining what they said as a whole (synthesis). In this step, disparate findings were systematically synthesized into a conceptual definition, reference architecture, and enabling technologies for ontology in MBSE, which were then consolidated to form a cohesive enabling framework.

(4) Grounding: Grounding bridged the theoretical synthesis with practical application, establishing a valid trajectory for the proposed enabling framework in future research and industrial contexts. This stage involved moving beyond the literature to validate the framework through the industrial applications.

3. State of the art and trends based on bibliometric analysis

3.1. Overview

A total of 566 publications related to ontology in MBSE (up to 2024) were collected. We adopted the following bibliometric metrics: (1) total publications (TP) per year; (2) total citations (TC) per year; and (3) average citations per publication (TC/TP) per year, to collectively reflect overall research productivity, academic impact, and publication quality. As illustrated in Fig. 2, the x-axis represents the publication year, while the y-axis denotes the TP in a given year. The color of a bubble represents the TC for a year, and its size indicates the TC/TP (owing to the small value of the TC/TP, the data for 2024 cannot be displayed).

As shown in Fig. 2, the TP showed an overall upward trend from 2008 to 2024, reflecting the growing interest and advancements in ontology and MBSE research. The emergence of related publications can be traced back to 2008. The impetus behind this was that in 2007, the INCOSE put forward the precise definition of MBSE in its publication "Systems Engineering Vision 2020" [32]. This seminal definition triggered a wave of interest. It emphasized the need for formalized semantics to support MBSE. Ontologies have emerged as a crucial semantic technology, providing a formal and machine-interpretable representation of domain knowledge that enables consistent system modeling. Consequently, research on ontology in MBSE gained momentum, which led to the first wave of related publications in 2008, marking the inception of a research paradigm that has evolved to date.

Moreover, the TC exhibited significant fluctuations in specific years, with notable peaks in 2011 and 2016, while the bubble in 2014 reached the largest size (the highest TC/TP ratio). In 2022, both TC and TP

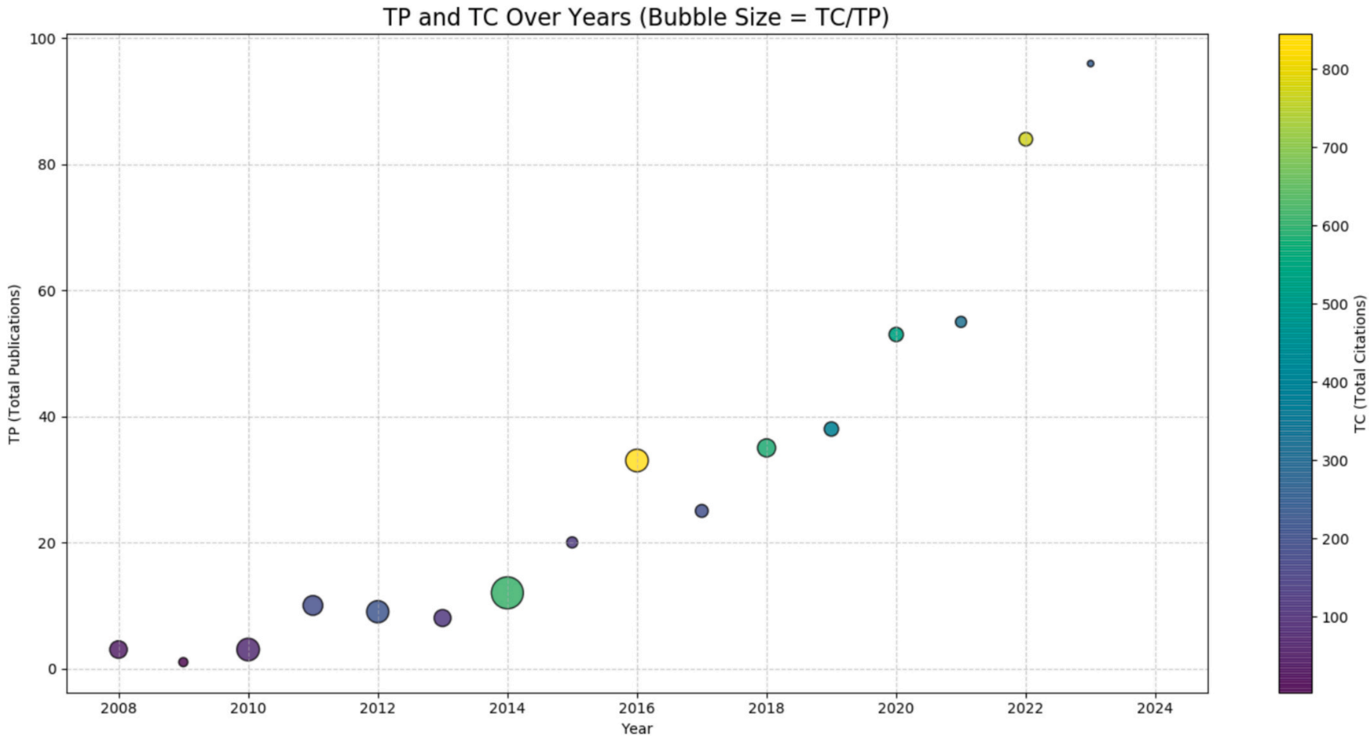

Fig. 2. Overview of publications.

increased markedly. These findings can be attributed to the crucial role of influential publications in shaping citation trends. The different editions of "A Practical Guide to SysML" [33] published in 2011, 2012, and 2014 significantly influenced trends, demonstrating its long-term impact on the research on ontology in MBSE. Notably, this book provides a practical guide to SysML and highlights that ontologies bridge formal semantics with SysML, facilitating knowledge sharing, enabling domain-specific reasoning, and improving interoperability in MBSE. The 2016 peak is largely attributed to "Design, Modelling, Simulation, and Integration of Cyber-Physical Systems: Methods and Applications" [34], published in *Computers in Industry*, which received 226 citations. It highlighted that ontologies have been developed to support CPS-design based on MBSE. More recently, "A Comprehensive Review of Digital Twin—Part 1: Modeling and Twinning Enabling Technologies" [35], published in *Structural and Multidisciplinary Optimization* in 2022, received 131 citations. In the context of a digital twin, ontology is used to uniquely refer to each possible object (i.e., component or sensor) in the physical asset, thereby supporting MBSE in establishing explicit relationships among them. This provides the digital twin with a unified and interpretable model that is essential for simulation, analysis, real-time monitoring, and predictive maintenance. These findings illustrate the evolution of ontology and MBSE research focus from SysML to cyber-physical systems (CPS) and, finally, digital twins, offering valuable insights into future research directions.

3.2. Research team analysis

Having examined the overall trends in publications and citations, we further performed co-authorship analysis to identify key research teams and their interrelationships in the research on ontology in MBSE. Nearly 10% of the total authors (1596) were included in the analysis by setting the minimum number of publications per author to three to ensure a clear and interpretable visualization. This filtering criterion resulted in the inclusion of 118 authors for the analysis. Using VOSViewer's clustering algorithm, closely related authors were grouped into clusters, and each cluster was assigned a unique color.

Fig. 3 illustrates the co-authorship network, showing 38 teams actively contributing to the field. The clustering analysis revealed that collaboration between teams was characterized by small-scale networks, predominantly led by core scholars. The most significant cluster, led by Lu, Jinzhi and Wang, Guoxin, included five interconnected teams, which exhibited extensive cross-team collaboration. They proposed a GOPPRR-E ontology for MBSE formalism, enabling the semantic integration of multiple MBSE modeling languages and domain-specific languages [21]. The second-largest cluster, headed by Dori, Dov, comprised three interconnected teams, demonstrating moderate collaboration. They introduced object process methodology (OPM) based on a minimal universal ontology of stateful objects and processes that transform them, which used two semantically equal formalisms: object-process diagram (OPD) and object process language (OPL), oriented toward humans as well as machines for MBSE modeling [36]. Other clusters comprised isolated teams. For instance, within the clusters highlighted by the black circles in Fig. 3, teams with nine, six, and five members indicated strong internal collaboration. The team led by Choley, Jean-Yves comprised nine members and primarily focused on leveraging ontologies to facilitate MBSE in the context of digital twins [37]. The group led by Kruse, Benjamin, with six members, concentrated on the integration of ontologies and MBSE in digital engineering, and proposed the digital engineering framework for integration and interoperability (DEFII) [38]. Meanwhile, the team led by Viehl, Alexander, comprising five members, emphasized the use of ontology to guide MBSE in enabling seamless collaboration among automotive manufacturers and suppliers [39]. However, the limited number of inter-cluster connections suggests that collaboration across research teams in the field of ontology in MBSE remains relatively fragmented, underscoring the importance of fostering stronger inter-team collaboration to advance the field.

To further explore collaboration patterns at a broader geographic scale, a bar chart that depicts the number of publications (N. of Documents) and collaboration patterns across different countries in the specific research domains is shown in Fig. 4. The chart employs two distinct bar colors to differentiate data types:

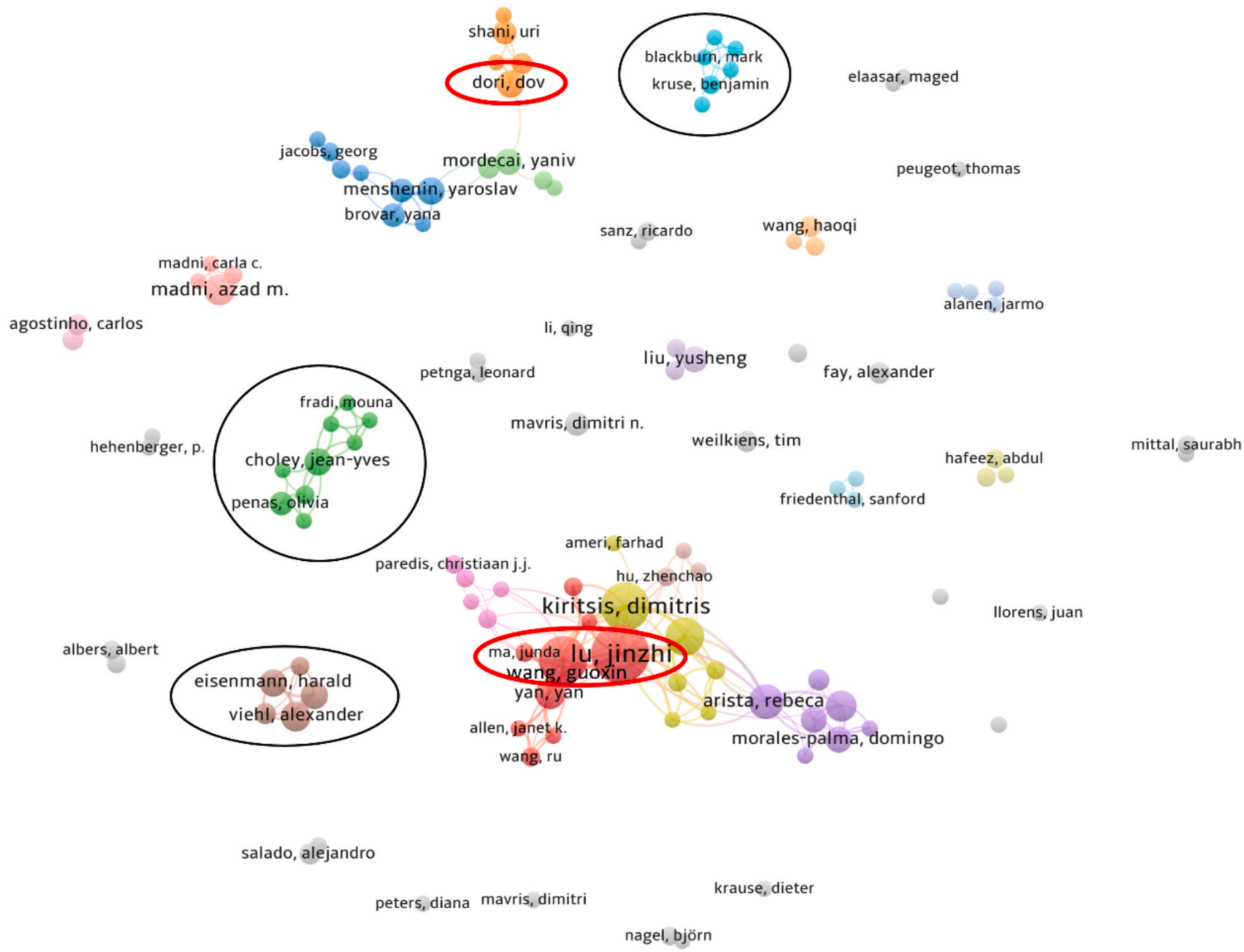

Fig. 3. Cluster network visualization.

- The red bars (multiple-country publications) represent the number of papers co-authored by authors from other countries. China has the longest red bar, indicating that China has the largest number of internationally co-authored publications in MBSE and ontology research. France follows closely, reflecting its notable contribution to international collaboration.
- The blue-green bars (single-country publications) represent the number of papers co-authored by authors from the same country. The USA has the longest blue-green bar, highlighting its leadership in independent publications. China also shows a notable blue-green bar, demonstrating its substantial independent research output in this field.

Accordingly, the USA ranks first in the cumulative total number of independent and collaborative publications. This can be attributed to several key factors. Organizations such as the National Aeronautics and Space Administration (NASA) have been at the forefront of adopting and advancing MBSE practices, driving the development of related research. NASA has developed a series of ontologies to support MBSE implementation in space systems, including the Trade Space Ontology (TSO), which was proposed to develop a crewed Moon to Mars Architecture and deal with a large decision space [40]. Moreover, the Common Space Systems Ontology (CoSSO) developed by the Advanced Concepts Office at NASA's Marshall Space Flight Center is used for modeling aerospace systems concepts in a pre-phase A context to support the transition to a more model-based paradigm [41]. China ranks second in the cumulative total number of independent and collaborative publications, a position bolstered by its robust ecosystem of academic societies, symposiums, and standardization efforts. Influential societies include the Systems Engineering Society of China, which is a leading organization promoting SE and MBSE research, hosting lecture series on emerging topics on semantic technologies in MBSE, especially ontology [42]. Notable symposiums and conferences, including the MBSE & Digital Engineering (DE) annual conference series [43], have increasingly emphasized the integration of formal ontologies into MBSE to enhance semantic interoperability and promote knowledge reuse across complex engineering systems. A significant milestone in China's MBSE standardization efforts is the recent release of GB/T 45803-2025 (Systems and software engineering-Unified Architecture Modeling Language for Model-based Systems Engineering) [44]. This standard implicitly emphasizes the alignment between formal modeling languages and ontology-based approaches, further reinforcing the foundational role of semantics in future MBSE development.

Germany, France, and Austria contribute a moderate number of independent and collaborative publications. Germany's Industry 4.0 initiative plays a crucial role [45], which seeks to connect the virtual and physical worlds through digitalization, with MBSE and ontology serving as key enablers for managing the complexity of modern industrial systems [46,47]. In addition, organizations such as the European Space Agency (ESA) [48], which includes Germany, France, and Austria as key

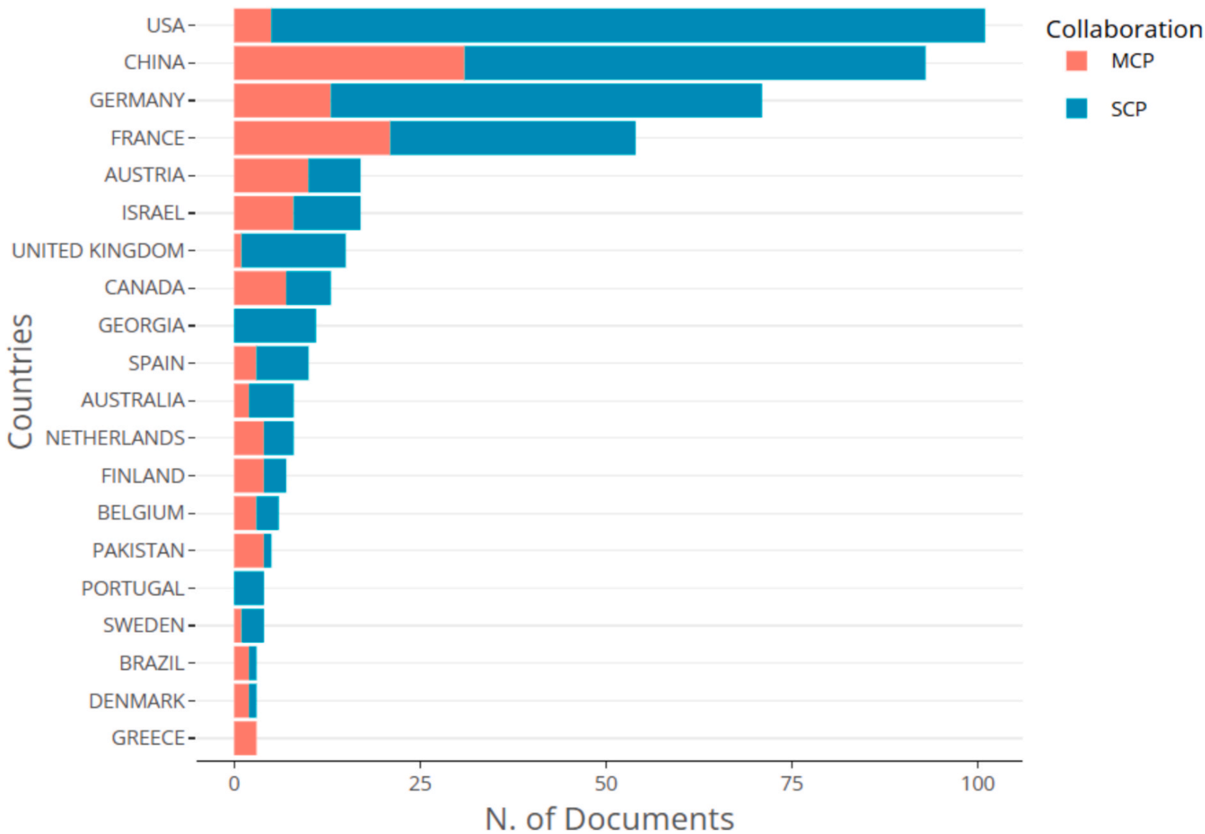

Fig. 4. Bar graph depicting the corresponding authors' countries versus the number of related publications (MCP: multiple-country publications; SCP: single-country publications).

member states, contribute significantly to the advancement of MBSE and ontology research. The ESA promotes the use of MBSE and ontology to effectively manage complex system design challenges, especially in aerospace and space exploration [49,50]. For example, Overall Semantic Modeling for Space System Engineering (OSMoSE) at the ESA introduces Space System Ontology to enable efficient semantic interoperability among model-based infrastructures [51]. Other countries such as Brazil, Denmark, and Greece show relatively lower publication output in this field. This may be due to varying levels of research focus, funding, and industrial demands for MBSE and ontology applications.

Furthermore, as shown in Fig. 5, the chord diagram reveals several salient patterns of international collaboration (minimum frequency = 4). The strongest link is observed between China and Switzerland (20), while China and Sweden (8) also demonstrate relatively close collaboration, underscoring China's intensive collaboration with European countries. France emerges as a central hub in the European network, with France–Spain (13) representing the second most frequent link, and maintaining regular partnerships with Austria (5), Italy (4), the Netherlands (5), Switzerland (6), and the United Kingdom (4), as well as Tunisia (5). In addition, Germany occupies a pivotal role, particularly through its collaborations with France (9), Switzerland (7), and the United Kingdom (4). The USA demonstrates diversified but moderate connections, most notably with Israel (7), as well as with China (5), Germany (4), Switzerland (4), and the United Kingdom (4). In contrast, Pakistan and Saudi Arabia (5), and Switzerland and Sweden (5) indicate more regionally confined cooperation. Taken together, the analyses indicate that Europe functions as the principal collaboration hub, with France, Germany, and Switzerland forming dense and interconnected clusters. China establishes strong bilateral ties with specific European countries, whereas the USA builds a broader but less intensive network. Beyond these core clusters, cross-regional collaboration remains limited, indicating the need for a more globally integrated research ecosystem in ontology and MBSE.

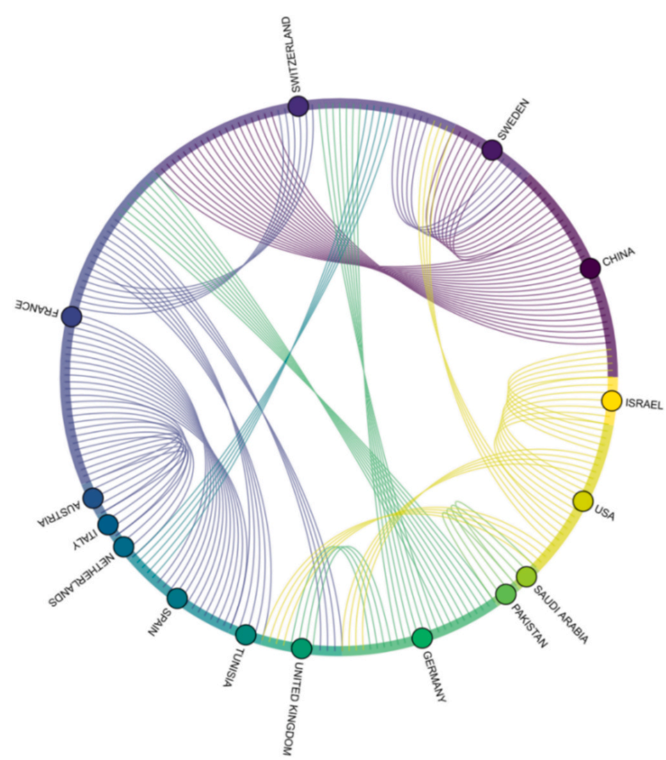

Fig. 5. Countries' collaboration patterns.

3.3. Source analysis

Following the analyses of research teams and their collaborative networks, we examined the primary publication sources that disseminate these cooperative outcomes. Fig. 6 shows the number of publications from the top 10 sources. Each data point is represented by a blue circle, with the number of publications labeled inside, and the intensity of the color increases with the publication number. Fig. 6 lists multiple sources, including academic journals, books, and conference proceedings. Based on their aims and scopes, they can be grouped into four primary domains: systems engineering, aerospace, industrial informatics, and engineering application, as follows:

(1) Systems engineering domain: ***Systems Engineering*** leads with 17 publications, underscoring its importance as the primary platform for MBSE and ontology research. This journal focuses on advancing systems thinking, model-based approaches, and integration techniques, making it a natural venue for foundational studies that explore how ontologies enhance semantic consistency, traceability, and interoperability in complex MBSE industry practice. ***Systems***, with 12 publications, concentrates on systems theory in practice, including fields such as systems engineering management and MBSE in complex industry systems, where ontologies provide semantic clarity and cross-domain integration. ***Handbook of MBSE***, with 10 publications, serves as a comprehensive reference on MBSE theory and practice. It includes chapters on ontology application because of its focus on standardization, model traceability, and the integration of diverse engineering models—all areas where ontologies are increasingly being adopted as enabling technologies;

(2) Aerospace domain: ***IEEE Aerospace Conference Proceedings***, with 16 publications, reflects the strong application of MBSE and ontology in high-assurance domains such as aerospace and defense. These fields demand rigorous system integration, life cycle management, and digital continuity, where ontology plays a critical role in enabling model interoperability across heterogeneous subsystems;

(3) Industrial informatics domain: ***IFIP Advances in Information and Communication Technology*** (14 publications), as a platform for cutting-edge research in ICT, this series frequently features work at the intersection of knowledge representation, enterprise systems, and intelligent integration. Ontology in MBSE is relevant here owing to its role in bridging engineering models with information technology systems for smarter automation and decision support. ***Advanced Engineering Informatics*** (9 publications) emphasizes the application of advanced informatics in engineering design. The research on ontology in MBSE fits well within its scope, especially for enabling knowledge-based modeling, intelligent reasoning, and semantic interoperability in engineering design. ***CEUR Workshop Proceedings*** (9 publications) is a prominent platform for early-stage and experimental research. The flexible and open-access format of CEUR makes it ideal for the rapid dissemination of research on ontology in MBSE. ***Journal of Industrial Information Integration*** (8 publications) is focused on the integration of industrial information systems across disciplines and organizational boundaries. Ontology in MBSE is particularly valuable here for facilitating semantic interoperability and data integration across the product life cycle, including design, production, and maintenance. The journal provides a suitable platform for publishing MBSE-related research that leverages ontologies for digital twin development, cross-domain system modeling, and standardized data exchange;

(4) Engineering application domain: ***Applied Sciences (Switzerland)*** (14 publications) supports multidisciplinary applied research and has become a key outlet for studies where ontologies are applied in MBSE to solve real-world engineering application challenges. ***Procedia CIRP*** (11 publications), as a leading series in the manufacturing and production engineering community, regularly features contributions related to digital manufacturing, Industry 4.0, and systems engineering. In these contexts, ontologies are integrated with MBSE to address semantic consistency, standards compliance, and system life cycle traceability, making this venue well suited for publishing practical engineering applications and case studies.

This distribution illustrates the diversity of research on ontology in MBSE, bridging theory with real-world applications and extending from domain-specific implementations to cross-disciplinary system integration. The publication sources indicate crucial platforms for conducting influential research and highlight evolving methodologies and emerging opportunities for further exploration.

3.4. Current research topics

Building on the insights obtained from source analysis, this study further investigated the thematic focus of research from these sources. A

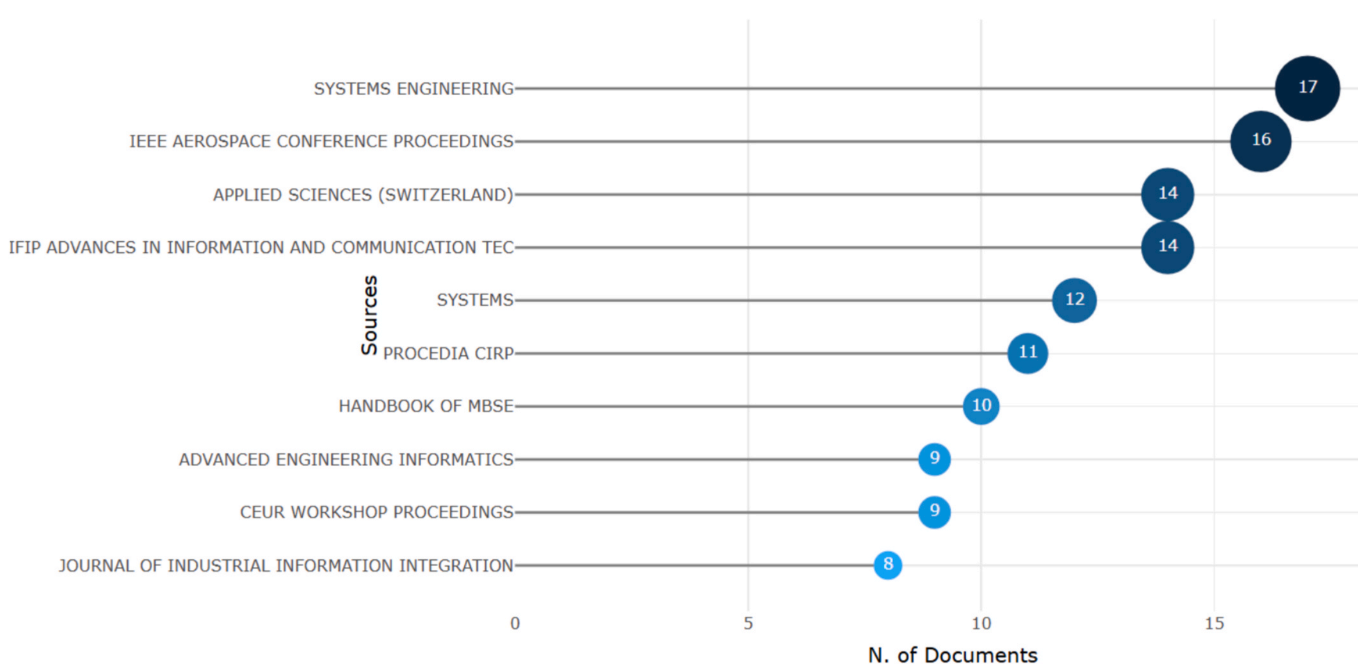

Fig. 6. Most relevant sources.

cluster analysis of co-occurrence keywords was performed using VOSviewer to identify current research topics. In VOSviewer, clusters are identified using a modularity-based community detection algorithm that maximizes intra-cluster similarity and minimizes inter-cluster connections, enabling the identification of coherent thematic clusters that reflect the underlying structure of the research field. Because *MBSE* and *ontology* were explicitly included as search keywords, they were more frequently found than any other related keywords (MBSE appeared 290 times; systems engineering, 178 times; and ontology, 133 times). These dominant terms occupied most of the network, overshadowing other nodes and making interpretation difficult. Thus, these three terms were removed from the initial data to improve data visualization. Fig. 7 depicts a co-occurrence network of authorial keywords in which the minimum threshold for the frequency of occurrence of keywords was set to 10. Thus, only those with at least 10 occurrences appeared in the network. Finally, 51 keywords were obtained (italicized hereafter for emphasis).

Five clusters were identified, as illustrated by the different colors in Fig. 7, where the size of each node indicated the frequency of keyword occurrences. Table 3 provides additional details about the clusters.

Cluster 1 (Red): Modeling Language

The topic of Cluster 1 is defined as "Modeling Language." In this cluster, *modeling languages* has the highest occurrence and the greatest total link strength, indicating its central role in the corresponding research topic. The remaining keywords in this cluster are closely related to this topic. *Semantics* constitutes a fundamental component of modeling languages [9]. Accordingly, semantically related concepts including the *semantic web* and *semantic modeling* are reasonably grouped within this cluster. Representative modeling languages such as *SysML* [7], *UML* [52], and *OWL* [53] naturally fall into the cluster. Moreover, MBSE modeling Languages play a critical role in supporting core systems engineering activities, including *requirements engineering* and *specification*, *system design*, *analysis*, as well as *verification and validation* [54], particularly in the context of *large-scale systems*. In addition, *model transformation* serves as a key technique for achieving interoperability between models expressed in different *modeling languages* [55].

Taken together, this cluster reveals that current research on modeling languages extends beyond the definition of modeling notations (e.g., *UML* and *SysML*) and increasingly emphasizes semantic foundations. The strong co-occurrence of semantics-related concepts (*semantics*, *semantic web*, *semantic modeling*, and *OWL*) indicates a growing focus on enabling semantic interoperability across heterogeneous modeling languages, rather than merely supporting syntactic-level *model transformation*. This shift has motivated the integration of semantic technologies, particularly ontology-based approaches, which constitute the core of the *semantic web*, into MBSE. Overall, this cluster reflects an evolution of modeling languages from purely descriptive representations toward semantically enriched and operational artifacts that support *requirements engineering* and *specification*, *system design*, *analysis*, as well as *verification and validation* activities in complex, large-scale systems engineering.

Cluster 2 (Green): Decision-Making and Knowledge Management

The topic of Cluster 2 is defined as "Decision-Making and Knowledge Management." In this cluster, *decision making* exhibits the highest occurrence and the greatest total link strength, while *knowledge management* ranks second in both, indicating that these two concepts constitute the core of the corresponding research topic. Their strong co-occurrence reflects a close and complementary relationship, in which *knowledge management* provides the structured knowledge resources and organizational mechanisms that underpin effective *decision-making* processes [56]. In this cluster, *knowledge management* is closely associated with *knowledge-based systems* and *knowledge graphs*, which serve as key technical means for representing, organizing, and reasoning over domain knowledge. *Knowledge graphs* build upon ontologies by instantiating them with real-world entities and facts, structured as interconnected nodes and edges that represent semantic relationships [57]. These knowledge-centric approaches enable the formalization and reuse of expertise and MBSE models, thereby supporting informed and consistent *decision-making* [58]. Correspondingly, d*ecision support systems* are naturally linked to *decision-making* because they operationalize

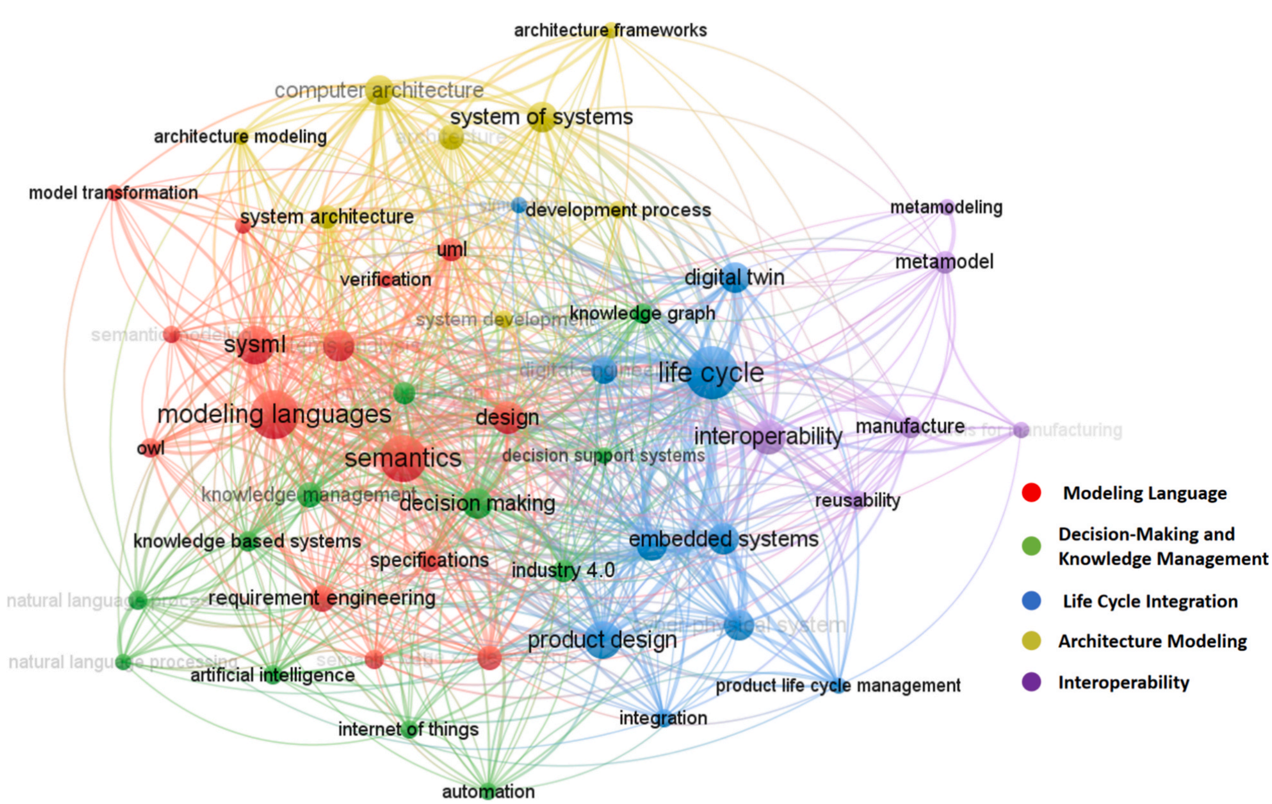

Fig. 7. Co-occurrence analysis (without MBSE, systems engineering, and ontology).

Table 3
Clusters of MBSE and ontology.

Cluster	Topic	Keyword	Occurrences	Total link strength	Total number
1	Modeling Language	modeling languages	87	281	15
		semantics	80	265	
		sysml	55	160	
		design	39	112	
		systems analysis	37	123	
		requirement engineering	27	66	
		large scale systems	22	65	
		uml	21	65	
		specifications	20	67	
		owl	15	48	
		semantic web	14	38	
		semantic modeling	12	48	
		verification	12	31	
		verification-and-validation	11	27	
		model transformation	11	38	
2	Decision-Making and Knowledge Management	decision making	34	128	12
		knowledge management	24	67	
		conceptual design	19	65	
		industry 4.0	19	45	
		knowledge-based systems	16	52	
		knowledge graph	16	54	
		artificial intelligence	15	49	
		natural language processing systems	15	58	
		internet of things	13	25	
		automation	12	35	
		decision support system	10	40	
		natural language processing	10	39	
3	Life Cycle Integration	life cycle	101	344	10
		product design	51	160	
		embedded systems	37	116	
		digital twin	35	95	
		information management	35	125	
		cyber-physical system	34	100	
		digital engineering	26	72	
		integration	13	39	
		product life cycle management	11	54	
		simulation	10	24	
4	Architecture Modeling	system of systems	35	83	8
		computer architecture	33	116	
		architecture	25	87	
		system architecture	21	77	
		system development	12	36	
		development process	12	27	
		architecture modeling	11	46	
		architecture frameworks	11	41	
5	Interoperability	interoperability	45	135	6
		metamodel	20	47	
		manufacture	19	71	
		reusability	15	54	
		models for manufacturing	11	25	
		metamodeling	10	22	

managed knowledge to assist both human-centered and automated decision processes.

With the rapid advancement of *artificial intelligence* (AI) [21], natural language processing (NLP) has increasingly enabled the automated extraction and structuring of knowledge from unstructured textual sources [59]. This capability significantly enhances the efficiency of knowledge acquisition, transforming raw data and textual information into actionable knowledge that can be directly exploited by *decision support systems*. Moreover, *conceptual design* represents an early and critical phase of system development in which *decision-making* is highly concentrated and knowledge-intensive, since designers must evaluate alternatives, constraints, and trade-offs under limited information [58]. With the background of *Industry 4.0* [60], *decision-making* is increasingly driven by large volumes of real-time data collected through the *Internet of Things* (IoT) [61]. However, data alone are often insufficient to support effective decisions in complex systems. Raw data lack explicit semantics, contextual relationships, and domain constraints, and data-driven models often suffer from limited interpretability and transferability. As reflected by this cluster, current research on MBSE demonstrates a trend toward knowledge- and data-driven decision-making, in which data are enriched, contextualized, and reasoned over using structured knowledge representations such as ontologies and *knowledge graphs*.

Cluster 3 (Blue): Life Cycle Integration

The topic of Cluster 3 is defined as “Life Cycle Integration.” Within this cluster, *life cycle* exhibits the highest occurrence and the greatest total link strength, underscoring its centrality in the research about ontology in MBSE. Its recurrence in compound terms such as *product life cycle management* further reinforces its semantic breadth, encompassing critical engineering phases represented by co-occurring terms such as *product design* and *simulation*. Within this context, concepts including *embedded systems*, *cyber-physical systems* (CPSs), *digital twins*, and *digital engineering* collectively reflect the increasing need to *integrate* heterogeneous models, data, and knowledge across the system *life cycle*. *Embedded systems* are resource-constrained computing units that are embedded within physical components, responsible for real-time sensing, actuation, and control [62]. *CPSs* form the backbone of digitalization by integrating physical processes with computational and

Table 4
Findings of SLR.

First author	TCs	Definitions	RA	Enabling technologies
Jacobs [85,86], Wang [87]	C1	modeling language definition	x	OSLC;/OML, RL
Arista [75]	C2	knowledge reuse	√	Metamodeling, OML, ontology
Cibrian [88], Medinacelli [89], Fu [90]	C2	knowledge acquisition, reuse, and refinement	x	KR;/OML, RL;/ metamodeling, KG, SR, KR
Chen [91,92], Alaoui [93]	C2	formalization/ verification	x	Ontology/OML, RL
Blasch [94]	C2	decision support	x	OML
Wagner [95]	C3	semantic integration	√	OML
Lu [21,96] Wang [97], Brovar [98]	C3	life cycle semantic integration	x	Metamodeling, OML;/ DSM
Feng [99], Laukotka [100], Ernadote [101], Madni [102]	C4	architecture modeling	x	Metamodeling/DT
Zindel [103], Debbech [104], Odukoya [105]	C4	architecture modeling	x	OML/ISO standard
Merrell [106], Orellana [107], Arena [108]	C5	semantic interoperability	x	Ontology, OML/ISO standard;/RL
Menshenin [109]	C5	semantic interoperability	x	DSM, dt
Yang [19], Medinacelli [110]	C1, C2	modeling language definition, knowledge sharing and reuse	x	OML
Hu[21,7596 76]	C1, C3	semantic integration	√	Ontology, metamodeling, OML
Lu [111]	C1, C5	formalization and semantic interoperability	x	Metamodeling, OML, OSLC
Arista [112], Novacek [113]	C2, C3	life cycle semantic integration, knowledge reuse, and decision support	x	Metamodeling;/OML, RL
Zheng [80]	C2, C3	semantic integration, knowledge reuse	√	Ontology, AI, GD
Zheng [79]	C2, C4	architecture modeling and trade-off analysis	x	Metamodeling, ontology, OML, KG, GD, CC
Dunbar [38]	C3, C5	semantic integration and interoperability.	√	OML, RL
Zhang [78]	C3, C5	semantic integration and interoperability.	x	Metamodeling, OML
Wu [5], Lu [114], Wang [115]	C2, C3, C5	formalization, life cycle semantic integration, and trade-off reasoning	x	Metamodeling, OML, OSLC, dt; ISO standard, OML, RL, KG, DT, IoT, cc, AI, ontology
Hu [69], Dong [116]	C2, C3, C5	formalization, semantic integration and interoperability, and knowledge reuse	x	DT, metamodeling, OML
Shani [117]	C1, C2, C4, C5	modeling language definition, semantic interoperability, architecture modeling, knowledge reuse	x	OML, OSLC

Notation:
The *Definition* column provides a concise summary of the aspects of MBSE that are supported by the application of ontologies, thereby contributing to the conceptual definition of OBMBSE.
TCs (Thematic Clusters): C1 = Modeling language; C2 = Decision-making and knowledge management; C3 = Life cycle integration; C4 = Architecture modeling; C5 = Interoperability.
Reference Architecture (RA): √ = Explicit reference architecture proposed; × = Not provided.
Enabling Technologies: OML = ontology modeling language; KG = knowledge graph; DSM = design structure matrix; GD = graph database; CC = cognitive computing; OSLC = Open Services for Lifecycle Collaboration; dt = digital thread; DT = digital twin; RL = semantic rule language; IoT = Internet of Things; cc = cloud computing; AI = artificial intelligence; KR = knowledge retrieval; SR = semantic reasoning.

communication capabilities [63]. The key challenges in engineering *embedded systems* and *CPSs* lie in the tight coupling of hardware and software, timing constraints, and integration with surrounding systems. MBSE offers precise models of physical structures, control logic, and interfaces, enabling early validation and consistency across hardware and software components. Ontologies complement this by formally defining the roles, functions, and communication semantics of embedded components (e.g., sensors and actuators), allowing for semantic *integration* across hardware and software domains. In the emerging paradigm of *digital engineering* (DE), *digital twins* further extend this *integration* by providing virtual replicas of physical assets or systems that enable real-time interaction between the digital and physical worlds, supporting system monitoring, optimization, and predictive analysis throughout the *product life cycle* [64–66]. Moreover, life cycle integration is achieved through effective *information management*.

Taken together, this cluster indicates that life cycle integration has emerged as a central research concern in the research on ontology in MBSE, driven by the need to maintain semantic consistency and information continuity across models, systems, and engineering phases. Rather than treating embedded systems, CPS, and digital twins as isolated artifacts, current research increasingly emphasizes ontology-based integration mechanisms that connect data, models, and knowledge across the entire system life cycle, thereby supporting more coherent, interoperable, and sustainable systems engineering processes.

Cluster 4 (Yellow): Architecture Modeling

The topic of Cluster 4 is defined as “Architecture Modeling.” The keywords grouped within this cluster, including *architecture*, *system architecture*, *computer architecture*, *architecture frameworks*, and *system of systems*, can be uniformly abstracted under the concept of *architecture modeling*. In the context of MBSE, *architecture modeling* constitutes a central activity in *system development* and the overall *development process*, since it provides a structured means to represent and analyze complex systems. Specifically, *architecture*, particularly *system architecture*, constitutes a fundamental aspect as it defines the structure, functions, interfaces, and behaviors of complex systems. *Computer architecture*, as a specialized subset, encompasses the structural organization of computing components (e.g., processors, memory, and buses). *Architectural modeling* serves as the essential activity [67,68], providing the structural and behavioral foundation for understanding system interactions, emergent behaviors, and operational constraints [69]. Ontologies provide a semantic foundation for both system and computer architectures by formalizing the definitions and relationships of architecture model elements [70,71]. Furthermore, MBSE extends beyond individual systems to *system of systems*, where multiple independent systems interact to fulfill broader mission objectives [72].

Taken together, this cluster indicates that the current research on architecture modeling in MBSE is moving beyond static structural representations toward semantically enriched and integrated architectural descriptions. The co-occurrence of *system architecture*, *computer architecture*, *architecture frameworks*, and *systems of systems* suggests an increasing emphasis on managing architectural complexity across multiple abstraction levels and system boundaries.

Cluster 5 (Purple): Interoperability

The topic of Cluster 5 is defined as “Interoperability.” In this cluster, *interoperability* exhibits the highest occurrence and the greatest total link strength, indicating that the ability to exchange, integrate, and reuse information across heterogeneous tools, models, and domains constitutes the central research focus. In this cluster, *metamodeling* plays a key role in promoting interoperability across various engineering domains by defining the MBSE modeling language used for model description and enabling its manipulation [73,74]. In this sense, *metamodels* function as a common structural contract that supports model exchange and transformation, which directly underpin *interoperability* and facilitate *reusability*. They empower the creation of standardized flexible MBSE models [5]. Fundamentally, *metamodels* are grounded in a shared ontology vision of the domain, providing explicit semantics for model constructs and relationships and thus supporting semantic interoperability beyond mere syntactic compatibility.

The presence of *manufacture* and *models for manufacturing* further indicates that *interoperability* is not treated solely as an abstract methodological concern, but also as a practical requirement in manufacturing-oriented contexts [75]. Manufacturing systems typically involve diverse life cycle tools (e.g., design, process planning, simulation, and monitoring) and multiple stakeholders, making *interoperability* essential for maintaining the continuity and consistency of information from engineering models to manufacturing models and shop-floor data. Consequently, research in this cluster emphasizes approaches that can bridge MBSE representations with manufacturing-specific models to support end-to-end interoperability and reuse. Taken together, this cluster suggests a research trajectory in which *interoperability* is increasingly achieved through the combined use of *metamodeling* and ontology.

3.5. Future trends

In this study, the trends of research topics on ontology in MBSE were analyzed using a thematic map. The horizontal axis in the thematic map represents the degree of relevance, indicating the centrality of a topic. The vertical axis represents the development density, reflecting the level of activity and development of a topic. The size of a circle on the thematic map represents the importance of a topic. A larger circle indicates a higher level of importance. As depicted in Fig. 8, the thematic map was divided into four quadrants based on the development degree (density) and the relevance degree (centrality).

Research topics in the first quadrant represent the motor theme, characterized by high centrality and density. These topics are significant and have garnered substantial interest. The life cycle integration cluster and the interoperability cluster are located in this quadrant. The life cycle interoperability and integration of MBSE models through ontologies have recently attracted marked interest [76–80]. This is driven by the growing demand for cross-domain collaboration, model continuity, and digital transformation in complex system engineering. Ontologies provide an integrated semantic representation across a system's life cycle, promote model interoperability, and ideally synchronize the use of terminology across stakeholders in MBSE design processes [70]. They are undergoing a notable evolution from domain ontologies toward upper (top-level) ontologies. While domain ontologies focus on capturing concepts and relationships specific to a particular application area (e.g., avionics, robotics, automotive), top-level ontologies such as Basic Formal Ontology (BFO), provide abstract, domain-independent constructs that serve as a semantic foundation for unifying diverse domains. This evolution reflects the increasing need for generalizable, interoperable, and life cycle semantic frameworks in MBSE.

The second quadrant represents niche themes, characterized by low centrality but high density, indicating well-developed yet specialized research topics. The decision-making and knowledge management cluster is located in this quadrant. Research on this topic is usually combined with data-driven technologies and algorithms. There has been growing interest in operations research (OR), and the machine learning (ML) community aims to combine prediction algorithms and optimization techniques to solve decision-making problems in the presence of uncertainty [81]. The research on data-driven technologies is mature enough, but it has less relevance to ontologies and MBSE. Moreover, the black-box nature, inherent biases in data or algorithms, and poor interpretability hinder data-driven decision-making. According to the keywords *decision-making* and *knowledge management,* knowledge combined data-driven decision-making in MBSE activities must be further investigated through ontologies and knowledge graphs.

The third quadrant is emerging or declining themes with low density and moderate centrality. The architecture modeling cluster is included in this quadrant. This is an emerging topic in the research of ontology in MBSE. This emergence is driven by the increasing complexity and inherent emergent properties of modern engineering systems, which necessitate a paradigmatic shift toward initiating abstract architecture modeling early in the design life cycle, preceding detailed design phases. However, current research on ontology in MBSE predominantly focuses on system architecture modeling, leveraging ontological reasoning to enhance consistency and traceability in system architecture design. Moreover, the characteristics of SoS such as interdependency, autonomy, and interoperability, pose significant challenges to the architecture

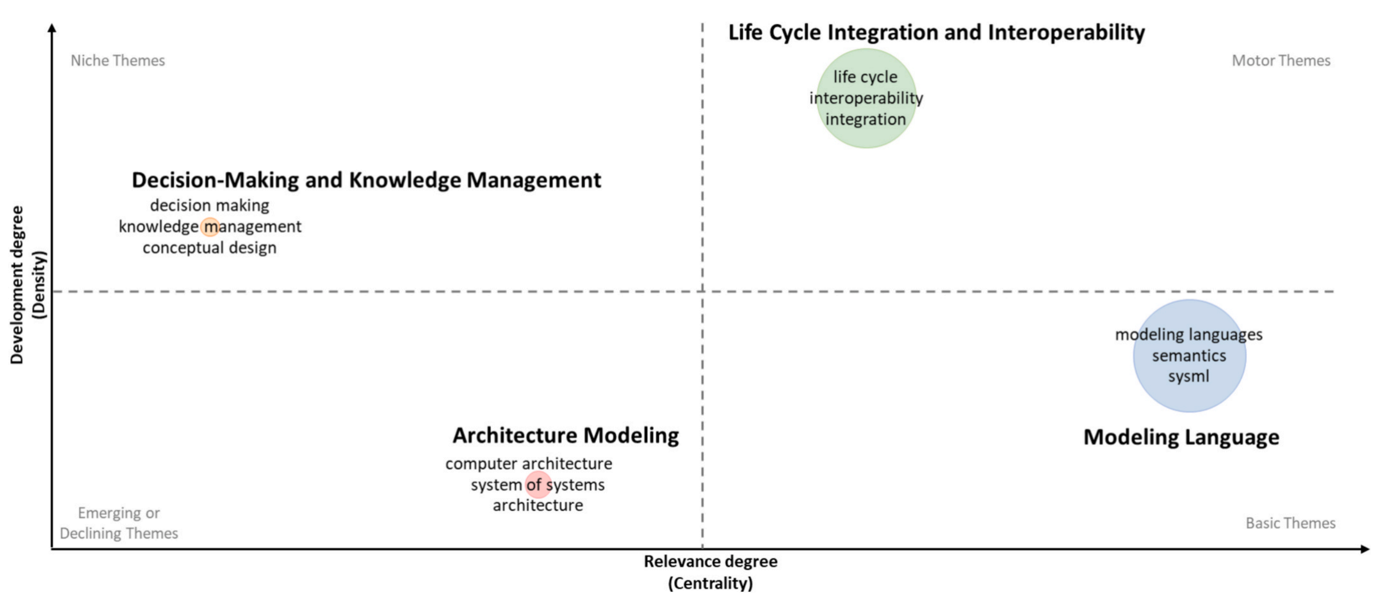

Fig. 8. Thematic map.

selection studies [82]. In the future, MBSE and ontologies must be used to support a broader range of architecture and SoS design schemes.

The fourth quadrant corresponds to the basic themes. The modeling language cluster is located in this quadrant, with high centrality and moderate density. This cluster represents a relatively weakly developed but important theme in the research of ontology in MBSE. Current modeling languages remain predominantly concentrated on first-generation graphical modeling languages, such as SysML. The emergence of this research reflects a movement toward semantic modeling languages, which use ontologies to define their syntax and semantics, such as SysML v2 and KARMA. SysML v2 is no longer merely a graphical modeling language. Rather, it encompasses multiple forms, such as text, graphs, and tables, all based on formal semantics [83]. Similarly, KARMA is a formal semantic modeling language based on the GOPPRR-E meta-metamodels [84].

4. Enabling framework based on the systematic literature review

4.1. Overview

The co-occurrence analysis presented in Section 3.4 delineates five dominant thematic clusters within the field of ontology in MBSE: (1) modeling language, (2) decision-making and knowledge management, (3) life cycle integration, (4) architecture modeling, and (5) interoperability. These five clusters constituted the search terms for the systematic literature review (SLR). By applying the selection criteria defined in Section 2.4, this study retained 52 primary studies for in-depth analysis, from an initial pool of 566 publications. These studies were systematically reviewed. The main findings are summarized in Table 4. These findings were synthesized to underpin the proposed ontology-based MBSE (OBMBSE) enabling framework comprising: (i) a conceptual definition, (ii) a reference architecture, and (iii) key enabling technologies. Furthermore, evidence from several research initiatives and pilot implementations indicated the practical effectiveness of the proposed OBMBSE framework. Overall, the SLR enabled a coherent synthesis that effectively bridged the macro-level thematic patterns revealed by bibliometric analysis with the micro-level empirical insights extracted from primary studies.

4.2. Definition of OBMBSE

Drawing on the evidence synthesized in Table 4, the SLR reveals that several recent research studies have contributed to the emergence of a new conceptual paradigm for ontology and MBSE. Across the identified studies, ontologies have been consistently positioned as more than auxiliary semantic annotations. Instead, they have been increasingly employed as foundational mechanisms that shape the formalization, integration, and operational use of MBSE artifacts throughout the system life cycle. These contributions can be systematically interpreted through the five thematic clusters identified in Section 3.4.

Specifically, ontologies are used to formalize the semantics of different ***modeling languages***, thereby guiding semantic integration across these languages. By providing unified, formal, and machine-interpretable semantics for representing architectural concepts, viewpoints, and relationships in ***architecture modeling***, ontologies drive consistent architecture descriptions and analysis. Simultaneously, ontologies can provide a common terminology for different knowledge bases, enabling knowledge alignment, which drives effective ***knowledge management,*** especially knowledge sharing and reuse [110]. Furthermore, ontologies can be used to represent design intent by formalizing goals, assumptions, constraints, rationales, and trade-offs as machine-interpretable semantics and linking them to architecture and verification artifacts, thereby enabling informed ***decision-making*** based on design axioms and reasoning rules [118119]. In addition, ontologies can standardize tool interfaces by aligning the terminology used in the input and output model data, ensuring semantic consistency during model information exchange [106], and thereby achieving semantic interoperability of heterogeneous models across tools and organizational boundaries. Moreover, throughout the system's life cycle, physical data generated during the operational phase can be fed back into digital MBSE models, influencing their design. By defining shared concepts and properties, and formalizing the relationships and constraints between MBSE models and physical data, ontologies enable ***life cycle integration*** and improve model effectiveness.

Based on the above synthesis, this paper formulates a conceptual definition of Ontology-Based Model-Based Systems Engineering (OBMBSE).

OBMBSE is a methodological and technical paradigm in which formal, machine-interpretable ontologies serve as the operational semantic core, encoding canonical vocabularies, relations, axioms, and inference rules for MBSE models and thereby achieving semantic integration and interoperability of models across domains, system levels, and the full system life cycle, as well as enabling data- and knowledge-driven decision support.

In this context, the ontology serves as the operational semantic core by explicitly formalizing modeling concepts and their relationships independently of specific tools or languages, using classes, relations, constraints, instances, and properties. Modeling constructs defined in different MBSE modeling languages and tools are formalized as ontology classes, while their structural constraints are represented as ontology relations and logical constraints on relations. MBSE models are represented as instances of the corresponding classes, and the interactions between model elements are captured as relationships subject to these semantic constraints. Additionally, the attributes of modeling constructs and models are formalized as ontology properties. This formalization enables a unified and machine-interpretable representation of domain knowledge across models. Furthermore, ontology-based rules support automated reasoning, for instance, if a requirement is allocated to a function and the function is realized by a component, then the requirement can be inferred to be satisfied by the component. Through such formalization and reasoning, the ontology enables executable semantics for interoperability, end-to-end traceability, and decision support. By grounding all modeling artifacts in a unified set of ontology classes, relations, and constraints, the ontology further enables the semantic integration and interoperability of models across engineering domains, system levels, and life cycle phases.

An example of vehicle automatic emergency braking (AEB) system design is further provided to illustrate how an OWL-based ontology can serve as an operational semantic core to represent knowledge and enable semantic integration across heterogeneous modeling languages. The AEB requirements, including *Collision Avoidance*, *Obstacle Detection*, *Collision Risk Assessment*, and *Automatic Braking*, are modeled using the <<requirement>> construct in the SysML Requirement Diagram, as shown in Fig. 9 (a). The <<requirement>> construct is formalized as the *Requirement* class, and an AEB requirement such as *Automatic Braking* is represented as the *Automatic Braking* individual of this class. Fig. 9 (f) shows that the property "ID: Req-1.3" of the *Automatic Braking* requirement is formalized as an annotation property "ID: Req-1.3" of the *Automatic Braking* individual.

Similarly, Fig. 9 (b) shows that the AEB functional features, including *Environment Perception*, *Collision Evaluation*, and *Brake Control*, are modeled using the modeling construct "VehicleFeature" in the EAST-ADL Feature Model. Fig. 9 (f) shows that the modeling construct "VehicleFeature" is formalized as the *VehicleFeature* class, and an AEB feature, such as *Brake Control*, is formalized as the *Brake Control* individual of this class.

Furthermore, the AEB behavioral processes, including events such as *Detect Obstacle*, *Evaluate Risk*, *Apply Brake*, and *Continue Driving*, are modeled using the modeling construct "Event" in the BPMN Process Diagram, as shown in Fig. 9 (c). The modeling construct "Event" is formalized as the *Event* class, and an AEB event such as *Apply Brake*, is represented as the *Apply Brake* individual of this class, as shown in Fig. 9

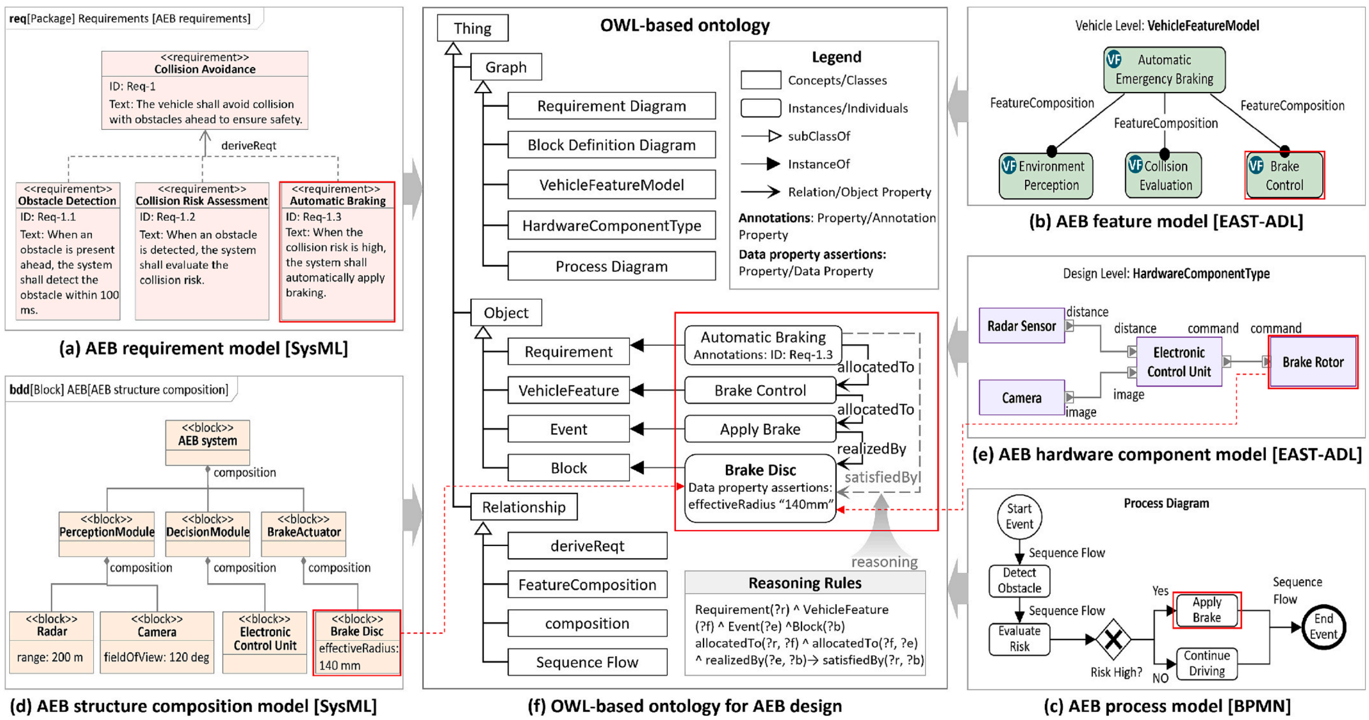

Fig. 9. Example of an OWL-based ontology integrating heterogeneous MBSE models from SysML, EAST-ADL, and BPMN for AEB system design.

(f).

In addition, the AEB structural components, including *Radar*, *Camera*, *Electronic Control Unit*, and *Brake Disc*, are modeled using the <<block>> construct in the SysML Block Definition Diagram, as shown in Fig. 9 (d). The <<block>> construct is formalized as the *Block* class, and an AEB system component such as *Brake Disc* is represented as the *Brake Disc* individual of the *Block* class. The attribute "effectiveRadius: 140 mm" of the *Brake Disc* is formalized as the data property "effectiveRadius "140 mm"" of the *Brake Disc* individual, as shown in Fig. 9 (f). The interactions of these hardware components are modeled using the EAST-ADL HardwareComponentType diagram, as shown in Fig. 9 (e). These hardware components are represented as *Radar Sensor*, *Camera*, *Electronic Control Unit*, and *Brake Rotor* through the modeling construct "HardwareComponentPrototype." In the ontology, the SysML "Block" and the EAST-ADL "HardwareComponentPrototype" are unified as the *Block* class to ensure cross-model semantic consistency. Furthermore, synonymous concepts are aligned during the integration process, for instance, "Radar Sensor" and "Radar" are unified as the *Radar Sensor* individual of the *Block* class; and "Brake Disc" and "Brake Rotor" are unified as the *Brake Disc* individual of the *Block* class, as shown in Fig. 9 (f).

To achieve semantic integration across these heterogeneous models, higher-level ontology concepts including *Graph*, *Object*, and *Relationship* are further abstracted. Based on this unified ontology, semantic connections among the *Automatic Braking* requirement, *Brake Control* feature, *Apply Brake* event, and *Brake Disc* component are established utilizing the object properties *allocatedTo* and *realizedBy*, as illustrated in Fig. 9 (f). Specifically, given the fact that the *Automatic Braking* requirement is allocated to the *Brake Control* feature, the *Brake Control* feature is allocated to the *Apply Brake* event, and the *Apply Brake* event is realized with the *Brake Disc* component, the ontology rules enable the automatic inference of the *satisfiedBy* relationship between the requirement and the component, thereby supporting informed decision-making.

The goals of this section are to establish a conceptually precise boundary for the OBMBSE and to provide a clear conceptual foundation for the reference architecture and enabling technologies presented in the subsequent sections.

4.3. Reference architecture of OBMBSE

A reference architecture is developed to facilitate a shared understanding across multiple products, organizations, and disciplines about current architecture(s) and future directions [120]. Through the SLR, several related reference architectures have been reviewed to guide the development of the reference architecture of the OBMBSE, as detailed below.

As shown in Fig. 10, Dunbar et al. [38] proposed a digital engineering framework for integration and interoperability (DEFII) comprising three layers: an ontology-aligned data layer that is tool- and domain-agnostic and supports interoperability, a reasoning layer that exploits automated reasoning capabilities provided by the Standard

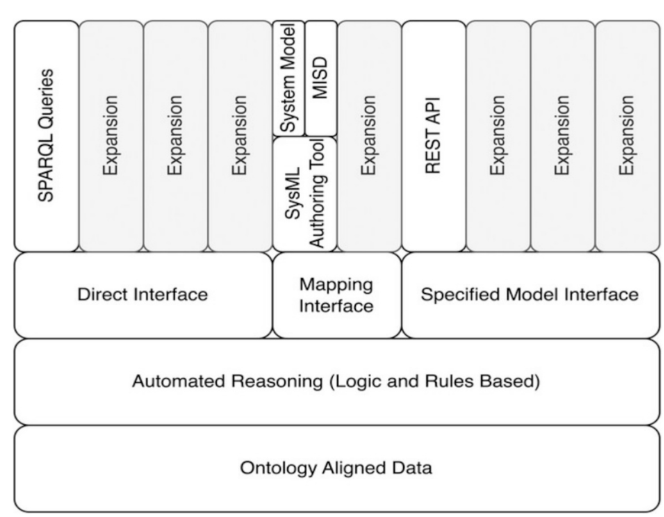

Fig. 10. Reference architecture of the DE domain space from Dunbar et al. [38].

Widget Toolkit (SWT) stack, and a notional interfaces layer that enables access to and mapping between MBSE or MBE models and ontology-aligned data. The notional interfaces include direct data access, ontology-based knowledge extraction, model-to-ontology mapping, and structured model exposure for analysis, supported by a reusable model interface specification diagram (MISD) based on the SysML parametric diagrams.

NASA Jet Propulsion Laboratory (JPL) introduced the Computer Aided Engineering for Spacecraft System Architectures Tool Suite (CEASAR) [95] as an MBSE analysis framework employing the Ontology Modeling Language (OML) to capture models using a controlled vocabulary and integrates heterogeneous design and analysis tools via adapters across multiple domains and life cycle phases, as shown in Fig. 11.

Arista et al. [75], Hu et al. [76], and Zheng et al. [80] proposed a reference architecture based on the Three-Layer Model (3LM), consisting of a data layer, an ontology layer, and a service layer, as shown in Fig. 12. The data layer aggregates heterogeneous data sources, including legacy and commercial software databases (e.g., PDM/CAD/CAM/ERP), cloud services, and data lakes, providing data, information, and knowledge to the ontology layer for semantic capture and instantiation. The ontology layer serves as the operational semantic core by integrating domain and model knowledge through upper-level and domain ontologies, enabling knowledge representation, reasoning, and semantic consistency across artifacts. Building on this semantic backbone, the service layer delivers MBSE-oriented functionalities, such as requirement management, architecture definition, verification, and visualization, while consuming and producing models and simulation results in a semantically coherent manner.

Lu et al. [21] proposed a reference architecture to integrate ontologies and MBSE models based on MOF and GOPPRR-E meta-meta-models, as illustrated in Fig. 13. This architecture supports various disciplines, such as control, hydraulics, and information, as well as different system engineering processes, including requirement, function, and architecture. In addition, it facilitates MBSE formalism across different stages of the product life cycle, including the concept and development stages.

Based on these related reference architectures, this study further proposes an OBMBSE reference architecture that aims to cover all the research topics identified in Section 3.4 and to guide the general MBSE scenarios that can be supported by OBMBSE, including modeling language definition, system architecture modeling, life cycle integration,

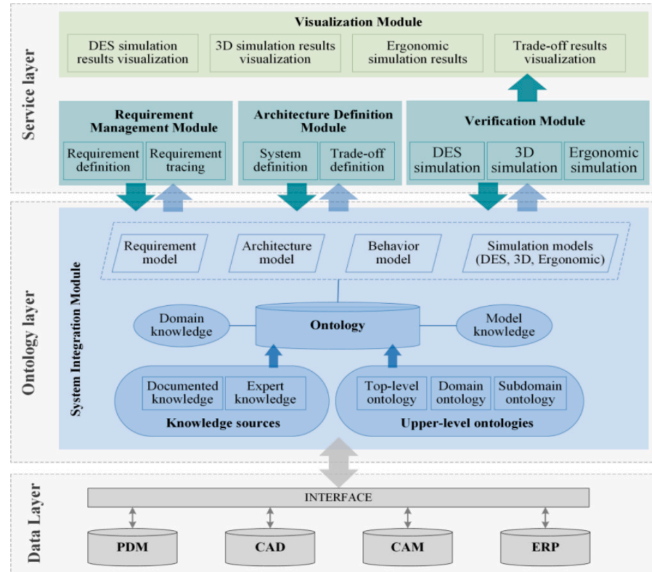

Fig. 12. A reference architecture based on the 3-Layer Model developed by Zheng et al. [80].

interoperability, knowledge management and decision-making. The proposed reference architecture is depicted in Fig. 14, and its main layers are as follows:

(1) OBMBSE modeling artifacts layer includes (i) modeling languages such as SysML, BPMN, and DSML, and (ii) architecture modeling activities covering SoS architecture modeling, system architecture modeling, and specialized domain-specific modeling to represent systems at multiple abstraction levels. The resulting models, which are produced through modeling languages and architecture modeling activities, are the core assets of MBSE and OBMBSE. These constitute the starting point of OBMBSE and the primary objects to be semantically governed. Through the OBMBSE reference architecture, these models can be grounded in explicit, machine-interpretable semantics, enabling consistent interpretation, cross-model traceability, and semantic-preserving exchange and analysis across tools and life cycle stages.

(2) OBMBSE reference architecture layer depicts an architecture that integrates MBSE practices with hierarchical ontologies based on MOF metamodeling principles. On the MBSE side, MOF meta-

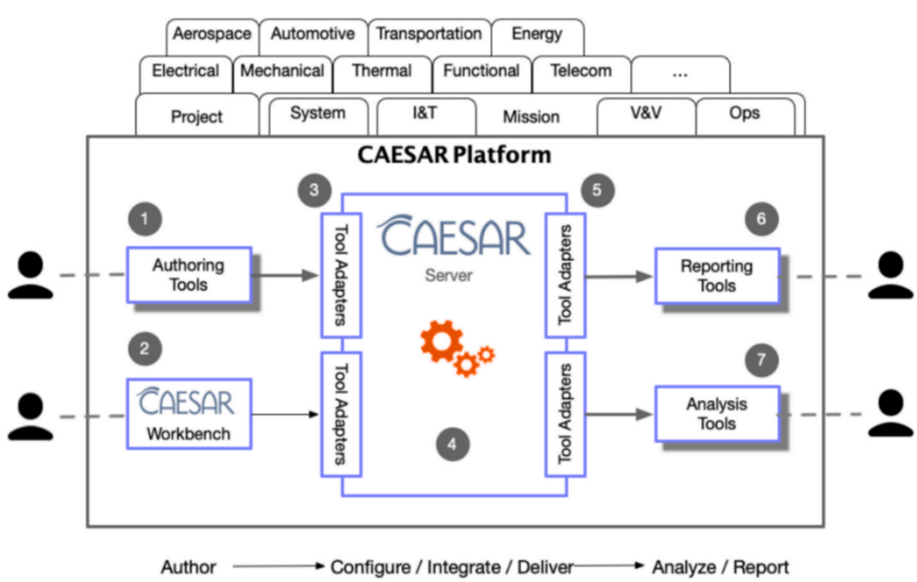

Fig. 11. CAESAR MBSE architecture from NASA [95].

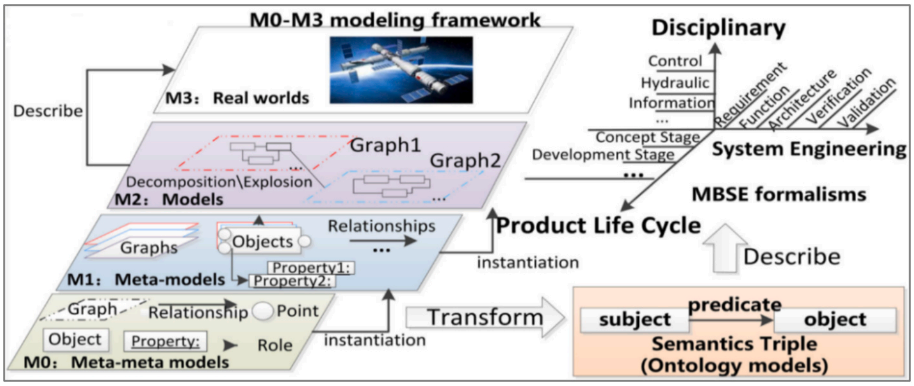

Fig. 13. Reference architecture of MBSE formalism developed by Lu et al. [21].

metamodels provide the foundational abstraction mechanisms for defining metamodels, which correspond to modeling languages used to construct system models. These modeling languages enable the systematic representation of the physical world. On the ontology side, the hierarchical ontologies incorporate a top-level ontology (e.g., BFO) and a mid-level ontology (e.g., IoF-Core) to ensure ontology-wide semantic coherence and interoperability. These upper ontologies further integrate domain ontologies that capture engineering domain knowledge. A mapping mechanism is established between MBSE artifacts and domain ontologies. Specifically, MBSE meta-metamodels are mapped to ontology concepts represented as classes, metamodels are mapped to domain concepts represented as subclasses, and MBSE models are mapped to domain instances represented as individuals. Through this alignment, MBSE models are semantically grounded in formal ontologies, enabling consistent interpretation and reasoning across modeling levels and engineering domains. The ontologies can further incorporate real engineering data from heterogeneous sources, such as design data from computer-aided design (CAD), manufacturing data from computer-aided manufacturing (CAM), analysis data from the finite element method (FEM), and simulation data from multibody system dynamics (MBS), which are formalized as instances of domain-specific classes or properties. Based on this, explicit semantic connections between MBSE model elements and real engineering data within a unified ontology can be established through ontology relations and properties. Moreover, ontology-based reasoning mechanisms support the inference of implicit semantic connections across models and data. Furthermore, a unified knowledge base is formed, comprising both a knowledge graph and a graph database, where MBSE model information and associated engineering data are instantiated, interconnected, and persistently stored. Moreover, the architecture integrates AI techniques. The ontologies and the knowledge base can support knowledge retrieval to enhance the interpretability, controllability, and reliability of AI-driven methods. Additionally, AI techniques can generate new knowledge artifacts, which further enrich and evolve the knowledge base. Together, they enable data- and knowledge-driven decision-making, thereby supporting the OBMBSE capabilities layer.

(3) OBMBSE capabilities layer includes unified modeling, interoperability, life cycle integration, knowledge management and decision-making capabilities. Specifically, unified modeling capability supports the unified modeling of different levels (e.g., SoS and system), domains (e.g., mechanical and hydraulic), and phases (e.g., requirement and architecture design) in a consistent manner. Interoperability capability enables consistent model information exchange based on a common ontology format across different modeling tools. The life cycle integration capability facilitates the seamless integration of physical data and digital models throughout the system life cycle, enabling continuous synchronization and traceability between real-world assets and their virtual representations. Knowledge management and decision-making capabilities leverage shared vocabularies, axioms, and reasoning mechanisms to support knowledge retrieval and reasoning. Furthermore, by integrating with data-driven approaches, such as machine learning, OBMBSE enhances informed engineering decision-making.

(4) OBMBSE industrial scenarios layer captures the application of OBMBSE in diverse industrial scenarios, including the following: **(i) system modeling:** OBMBSE enables modeling across different modeling languages guided by hierarchical ontologies that provide unified semantics; **(ii) model information exchange across different tools:** OBMBSE supports seamless exchange of model information among tools such as Enterprise Architect and MetaGraph, which typically utilize standardized formats such as .owl files to ensure interoperability; **(iii) life cycle integration of digital models and physical data:** OBMBSE integrates life cycle digital models and physical data into a unified ontology. For instance, in the metal forming process, the architecture models representing the metal forming process, Petri Net models capturing process dynamics, and corresponding physical data are integrated into an ontology to enable holistic analysis and traceability; **(iv) design alternatives trade-off analysis and selection:** OBMBSE formalizes design alternatives as ontologies and defines feasible design boundaries through ontology rules. Utilizing ontology-based reasoning, the admissible design space is derived. Within this constrained space, an AI-assisted multi-objective decision-making algorithm can jointly consider design objectives and constraints, explore the expected solution space, and identify the optimal design solution by trading off multiple objectives.

The reference architecture provides guidance for the implementation and deployment of OBMBSE in industrial applications. However, it also requires many advanced technologies to achieve the OBMBSE capabilities, as discussed in Section 4.4.

4.4. Key enabling technologies for OBMBSE

The *Enabling technologies* column in Table 4 summarizes the techniques adopted across the reviewed studies. To synthesize these findings, this study organizes the enabling technologies according to the research topics identified in Section 3.4, as shown in Fig. 15.

Modeling language: Ontologies introduce formal, machine-interpretable semantics to MBSE modeling languages, thereby achieving the semantic integration of heterogeneous modeling languages. Key technologies supporting this capability include metamodeling technologies, ontology modeling languages, and semantic rule languages. Metamodeling technologies, such as MOF and GOPPRR-based frameworks [21], provide structured definitions of modeling languages and their abstract syntax, thereby facilitating the generation, transformation, and alignment of model elements across multiple modeling languages. Ontology modeling languages, such as OWL, are grounded in description logics, which mitigates the lack of formal semantics in conventional MBSE modeling languages [19]. Ontology Modeling Language (OML) provides a tool-neutral language for

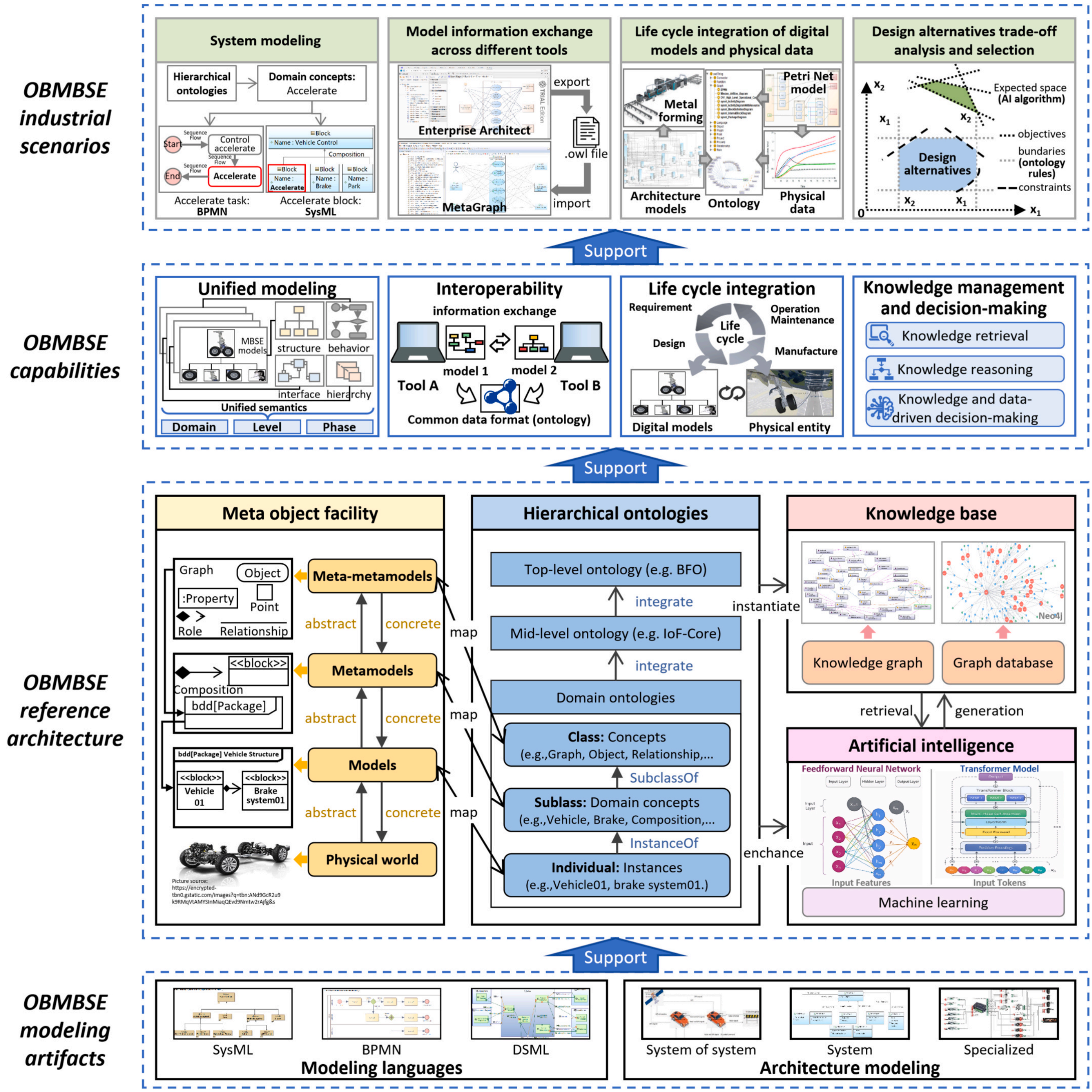

Fig. 14. Reference architecture for OBMBSE.

representing, exchanging, and analyzing systems engineering information. As a restricted, pattern-based profile aligned with OWL2-DL, it provides semantic-web reasoning capabilities for MBSE workflows [89,95,103,105]. Semantic rule languages, such as W3C's SHACL, enable constraint-based checks (e.g., well-formedness and consistency) directly over OWL representations [38].

Architecture modeling: Architecture modeling constitutes a core research topic in OBMBSE, focusing on the formal representation and analysis of system architectures. Key enabling technologies include metamodeling technologies, ontology modeling language, and ISO standards. Metamodeling technologies such as MOF [5,21,75,79,96,97,111,116] and architecture-oriented metamodeling patterns (e.g., ARIS, Ecore, GME, and GOPPRRR [121]) provide a structured approach for defining architectural elements, relationships, and constraints. They also enable model transformation and support tool interoperability, facilitating consistent and automated architecture modeling workflows. Ontology modeling languages, such as OWL and OML, drive metamodeling by formalizing the semantics of architectural concepts and relationships [105]. They drive semantic integration across different modeling languages and viewpoints, enabling rigorous representation, reasoning, and verification of architecture models. By providing machine-interpretable definitions for architectural elements and their interrelationships, ontology modeling languages support traceability, consistency checks, and automated reasoning within MBSE workflows. Moreover, ISO standards, such as ISO/IEC/IEEE 42010:2011 [105], standardize the notion of architecture description, providing a common basis for representing viewpoints, concerns, stakeholders, and architectural rationale.

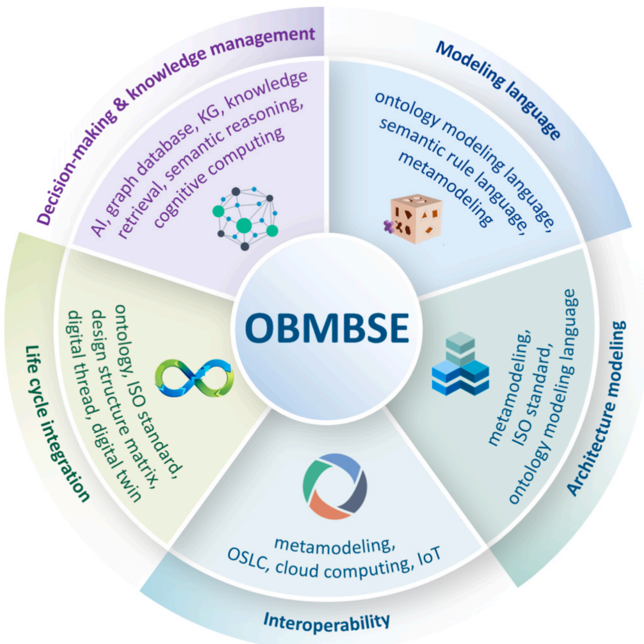

Fig. 15. Key enabling technologies of OBMBSE.

Decision-making and knowledge management: Recent research has exhibited a growing trend toward combining knowledge-driven methods with data-driven techniques to enhance decision support and knowledge management capabilities. Key enabling technologies include graph databases, knowledge graphs (KGs), knowledge retrieval, semantic reasoning, AI, and cognitive computing. A common implementation pattern is to store ontology-instantiated knowledge in a graph database, such as Neo4j, often hosted in the cloud to enable distributed access across partners [79,80], thereby supporting collaborative knowledge management. Moreover, KGs represent information in a triple format with entities and relationships grounded in the defined ontology, enabling structured model knowledge representation, integration, and reuse across heterogeneous engineering domains [90,114,115]. Building upon KGs, knowledge retrieval technologies, such as graph and pattern matching, support model reuse by enabling the identification, retrieval, and adaptation of relevant model elements, relationships, and structural patterns across heterogeneous engineering models [88]. Semantic reasoning technologies, such as ontology reasoning and rule-based inference (e.g., description logic reasoning and rule engines), further support reasoning-enabled analytics and decision-making by deriving implicit knowledge, detecting inconsistencies, and validating design assumptions [89]. In parallel, data-driven technologies, especially AI technologies, such as machine learning (ML) models have been increasingly incorporated to enhance trade-off analysis and reasoning capabilities [114]. ML techniques, such as natural language processing (NLP) technology and graph computing, provide possible solutions to promote knowledge acquisition and reuse [80]. With the development of AI algorithms, cognitive computing [79] has simulated human-like understanding, reasoning, and learning capabilities to support intelligent decision-making.

Life cycle integration: To enable higher-level intelligence, it is necessary to achieve the integration of the model, data, information, and knowledge about the system for the entire life cycle [79]. Key enabling technologies include ontologies, ISO standards, design structure matrix (DSM), digital thread, and digital twin. Ontologies, especially top-level ontologies, including the descriptive ontology for linguistic and cognitive engineering (DOLCE) [114], BFO [69,75,91,107,115], and the mid-level IOF-Core ontology [75,76,79,80], provide stable and abstract semantic frameworks and long-term governance mechanisms, which support semantic integration among MBSE models across different life cycle stages. The ISO standards, such as ISO/IEC 21838-1, specify the required characteristics of a domain-neutral top-level ontology that can be used in tandem with domain ontologies at lower levels to support the integration of life cycle models [122]. ISO/IEC/IEEE 15288 establishes a common framework of process descriptions to describe the life cycle of system models created by humans, defining a set of processes and associated terminology from an engineering viewpoint [107]. In addition, DSM-based approaches [98,109] map the core entities from conceptual to detailed design and to manufacturing stages, thereby supporting the management of interdependencies across the life cycle. With the advances in digital engineering, the digital thread enables life cycle integration in OBMBSE via the continuous linking and traceability of models, data, and decisions across life cycle phases, supporting lifecycle-wide information continuity and traceability in MBSE environments [5]. Moreover, by synchronizing the states of physical and digital systems, digital twins enable lifecycle-aware analysis and the integration of design, manufacturing, and operational knowledge [102].

Interoperability: Interoperability focuses on enabling heterogeneous models across tools to exchange information and correctly interpret its meaning. Key enabling technologies include metamodeling, Open Services for Lifecycle Collaboration (OSLC), cloud computing, and IoT. Metamodeling technologies provide a unified syntactic foundation for aligning heterogeneous model structures and supporting model transformation across tools [111]. OSLC [5,85,86,111,117] provides a practical mechanism for linking heterogeneous artifacts across engineering tools through standardized, RESTful interfaces. By enabling tool-level connectivity without requiring full data integration, OSLC supports lifecycle-wide semantic interoperability while preserving tool autonomy. Cloud computing [114] further supports interoperability by providing a scalable, distributed infrastructure for hosting interoperable services, shared data repositories, and collaboration platforms. By enabling ubiquitous access and integration across geographically distributed stakeholders and tools, cloud-based environments facilitate cross-organizational interoperability in OBMBSE contexts. In addition, IoT enables interoperability in OBMBSE by the real-time acquisition and synchronization of operational data from physical systems with digital models [114].

4.5. Industrial applications

The proposed OBMBSE enabling framework has been examined using selected applications in complex industrial systems, providing illustrative and preliminary evidence of its potential applicability and relevance in an industrial context. Because internal and external validity are widely recognized as fundamental criteria for evaluating the effectiveness of frameworks [123], they were selected for evaluation in this study. Specifically, the internal validity indicates the extent to which an industrial application establishes the cause-and-effect relationship between the proposed OBMBSE enabling framework and the observed outcome [124]. In this context, the internal validity is reflected in the framework's capabilities to support unified modeling, interoperability, life cycle integration, knowledge management and decision-making in industrial applications. Meanwhile, the external validity shows the extent to which the proposed framework can be generalized and replicated across different engineering projects and domains [124]. A series of research initiatives and pilot deployments have preliminarily examined the potential internal and external validity of the proposed OBMBSE enabling framework across diverse real-world industrial scenarios.

Within the completed European Union (EU) H2020 project QU4LITY (825030), the OBMBSE framework was applied to the manufacturing system design for the Airbus A320-XLR (eXtra Long Range), with a particular focus on the fuselage orbital joint process [75]. The analysis of this type of manufacturing system led to an exponential number of alternative solutions being analyzed, eventually reaching the limits of the design space and becoming unsolvable. To address this challenge, the OBMBSE framework was used to support automatic orbital joint process design and to perform trade-offs according to the needs of different stakeholders. The framework enabled unified modeling of mission views, operational scenarios, functional views, logical views, and physical structures of the assembly processes, based on the GOPPRR-E meta-metamodels. Based on this, an application ontology was developed based on mid-level IOF-Core ontology and top-level BFO to enable the semantic integration and interoperability of physical assembly system domain knowledge (e.g., assembly process specifications, material, resource, costs, process), industrial requirements, and virtual MBSE architecture models throughout the life cycle phases of the Airbus A320-XLR. As illustrated in Fig. 16, the implementation of OBMBSE followed the three-level mappings defined in the reference architecture: the GOPPRR-E meta-metamodels were mapped to OWL classes, the SysML and simulation metamodels were mapped to OWL subclasses, and the MBSE architecture models and behavioral simulation models were mapped to OWL individuals, yielding a knowledge graph for knowledge management. Leveraging this knowledge graph, a generative algorithm automatically produced 18 candidate orbital joint process designs. Furthermore, the framework enabled the automatic synthesis of simulation models from architecture models using extended design rules and constraints, allowing alternative process sequences to be simulated and their performance quantitatively compared. Therefore, the proposed framework could support the selection of the optimal design solution for the fuselage orbital joint process in the A320-XLR manufacturing system design. This industrial application provided preliminary insights into the potential internal validity of the proposed OBMBSE enabling framework, which was reflected in its capability to support unified modeling, interoperability, life cycle integration, and knowledge management and decision-making in the design of the fuselage orbital joint process for the Airbus A320-XLR.

In another completed EU H2020 project entitled FACTLOG (869951), the OBMBSE framework was further applied in a real industrial scenario at BRC Ltd (the UK's largest supplier of steel reinforcement and associated products), focusing on a specific bay of one of its steel reinforcing installations. Three general product families were produced: Coils, Bars, and Bent Bars. Coils and Bars were produced after a one-stage process, while the Bent Bars required a two-stage process. The machines performing a type of process were not all identical, since they might have different constraints and production characteristics because of different levels of automation, operator experience, or working parameters. Consequently, it was challenging to seamlessly integrate the information from different life cycle phases and address the concerns of different stakeholders. To address this challenge, Lu et al. [96] proposed a semantic Model-Based Systems Engineering (sMBSE) approach as a concrete implementation of the OBMBSE framework. As shown in Fig. 17, the GOPPRR-E meta-metamodels were utilized to define domain-specific metamodels covering five key system perspectives: mission, operation, function, logic, and physical structure, thereby enabling unified modeling of the architecture models of the metal forming process in the real world. These modeling constructs (meta-metamodels, metamodels, and models) across the M3, M2, and M1 layers were formally defined in an OWL-based ontology using classes, object properties, subclasses, and individuals. This enabled the metal forming process to be defined based on the unified ontology, which promoted data interoperability across the entire design process. Leveraging this ontology, Petri Net (PN) simulation models representing the number and sizes of orders, as well as their assignment to specific machines, were automatically generated from the MBSE architecture models, ensuring semantic consistency across modeling domains. The automated simulation supported decision-making to promote the design efficiency of the metal forming process. This industrial application provides preliminary evidence for discussing the potential internal validity of the proposed OBMBSE enabling framework, particularly regarding its capability to support unified modeling and enable interoperability and data-driven decision-making in the design of the metal forming process.

Moreover, in the space industry, the ongoing Model Based 4(for) System Engineering (MB4SE) initiative [125], led by ESA, aims to federate the system engineering community toward a common ontology-based modeling paradigm. The engineering of space systems is a collaborative, iterative process that integrates various domain-specific viewpoints to represent the final system. To ensure consistency across these viewpoints, the Overall Semantic Modeling for System Engineering (OSMoSE) at ESA, introduced a top-level ontology called Space System Ontology (SSO) to enable efficient semantic interoperability and integration among model-based infrastructures across collaborative space system developments [51].

These projects and initiatives combined provide illustrative evidence of the external validity of the proposed OBMBSE enabling framework, demonstrating its potential applicability and generalizability across diverse industrial contexts, including civil aviation manufacturing, metal forming processes, and space systems engineering.

In addition, some commercial MBSE modeling tools contribute to the practical implementation and industrial uptake of the OBMBSE enabling framework. For instance, Enterprise Architect [126] supports the development of large-scale ontologies within a fully-integrated MBSE modeling environment. It supports the bidirectional transformation between MBSE models and OWL/RDF ontologies, enabling the integration of system architecture models with formal semantic representations. By embedding ontologies directly into the MBSE workflow,

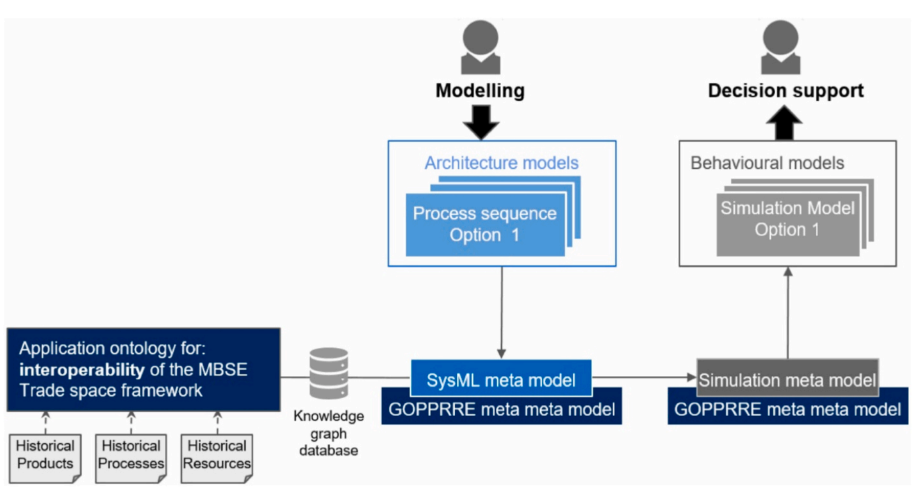

Fig. 16. Functional diagram of the interoperable MBSE tradespace system from Arista et al. [75].

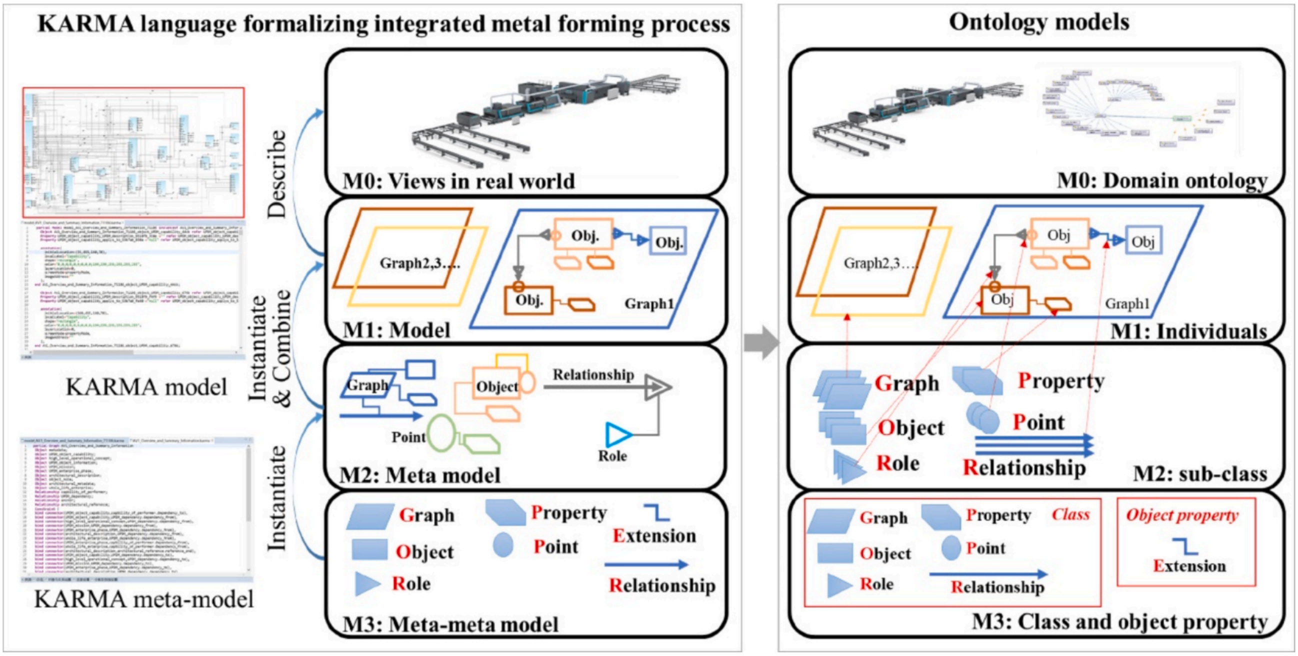

Fig. 17. M0-M3 modeling framework based on KARMA from Lu et al. [96].

these tools can help operationalize the OBMBSE framework in industrial contexts, providing a concrete mechanism for its adoption.

In summary, these industrial applications provide illustrative and preliminary evidence for discussing the potential internal and external validity of the proposed OBMBSE enabling framework. However, the development and maintenance of ontologies can be time-consuming and require substantial domain expertise, and the associated benefits may not be immediately realized. Given these considerations, the proposed framework is particularly applicable to large-scale, complex, and collaborative engineering domains. These domains are characterized by high complexity, strong multidisciplinary coupling, distributed stakeholders, and lifecycle-oriented development processes, where effective integration and interoperability across industrial disciplines and life cycle stages are essential for ensuring system consistency and coordination. In such contexts, the initial investment can be justified by long-term gains in reduced integration effort, improved traceability, and enhanced decision support. Nevertheless, it should be noted that the current evidence is primarily qualitative, and quantitative benchmarking of the proposed framework's effectiveness remains limited.

5. Conclusion

This study conducts a systematic bibliometric analysis combined with an SLR of 566 publications from 2008 to 2024 in the field of ontology in MBSE, using VOSviewer and BibliometriX. The findings reveal an overall upward trend in research activity. At least 39 active research teams are identified, characterized by small-scale collaboration networks centered around core scholars. Geographically, the research landscape exhibits a tiered structure. Moreover, the ten most prolific sources, grouped into systems engineering, aerospace, industrial informatics, and engineering application, highlight the field's interdisciplinary nature and blend of theory and practice. Furthermore, the analyses reveal the five key features (modeling language, decision-making and knowledge management, life cycle integration, architecture modeling, and interoperability) and future trends of ontology in MBSE. In the future, research in this field is expected to increasingly focus on life cycle integration and interoperability, while further advancing data- and knowledge-driven decision-making. The scope of research is also expected to broaden, extending from system architecture toward SoS architecture design. In addition, modeling languages are evolving toward semantic modeling languages, which use ontologies to define their syntax and semantics, such as SysML v2 and KARMA.

It should be noted that, unlike prior reviews that have not systematically examined the state of the art or provided an enabling framework for integrating ontologies with MBSE, this study presents a more structured and integrated discussion of ontology in MBSE and further proposes an OBMBSE enabling framework, including an explicit definition, a reference architecture, and enabling technologies.

Furthermore, the potential industrial applicability of the proposed OBMBSE enabling framework has been discussed for diverse sectors, including Airbus A320-XLR fuselage orbital joint design and steel reinforcement metal forming process design at BRC Ltd. Beyond these specific cases, under the ESA-led MB4SE initiative, OSMoSE introduces a top-level ontology SSO to facilitate efficient semantic interoperability and integration across model-based infrastructures in collaborative space system development. Moreover, commercial MBSE modeling tools can further support the practical deployment and industrial adoption of the OBMBSE framework.

However, as an exploratory study, this study has several limitations. First, the review is limited to the Scopus database, which, despite its breadth, may omit relevant publications such as industrial standards, technical reports, and white papers documenting practical implementations. Second, the implementation of the OBMBSE framework inherently depends on the collaboration of multiple enabling technologies. This study could not comprehensively cover all such enabling technologies in detail. Third, developing and maintaining ontologies can be time-consuming and require substantial domain expertise, which may limit their scalability and practical adoption. In addition, the benefits of the proposed OBMBSE enabling framework may not be immediately realized, since they often depend on long-term integration and reuse across the system life cycle. Fourth, the potential effectiveness of the proposed framework is preliminarily assessed in a qualitative manner based on the selected industrial applications, but this study lacks a comprehensive set of quantitative benchmarking metrics for systematic evaluation of the return on investment and performance improvements associated with the framework. Fifth, although the presented industrial applications indicate practical potential, they remain relatively limited, and broader adoption across diverse engineering domains is yet to be observed.

Therefore, future research will further expand the literature base beyond academic publications to include industrial white papers, standards, and commercial implementations, providing a more complete understanding of OBMBSE in practice. In parallel, an ongoing investigation into enabling technologies remains necessary to fully realize the proposed OBMBSE framework. Moreover, the development of standardized evaluation metrics and benchmarking methods is essential to quantitatively assess the effectiveness and performance of the proposed framework. In addition, further empirical studies are required to explore the proposed framework's deployment in complex, large-scale, and cross-domain industrial systems design.

CRediT authorship contribution statement

Mengru Dong: Writing – original draft, Visualization, Methodology, Data curation, Conceptualization. **Guoxin Wang:** Writing – review & editing, Methodology. **Jinzhi Lu:** Writing – review & editing. **Shouxuan Wu:** Writing – review & editing. **Yihui Gong:** Methodology. **Yan Yan:** Methodology, Conceptualization. **Dimitris Kiritsis:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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